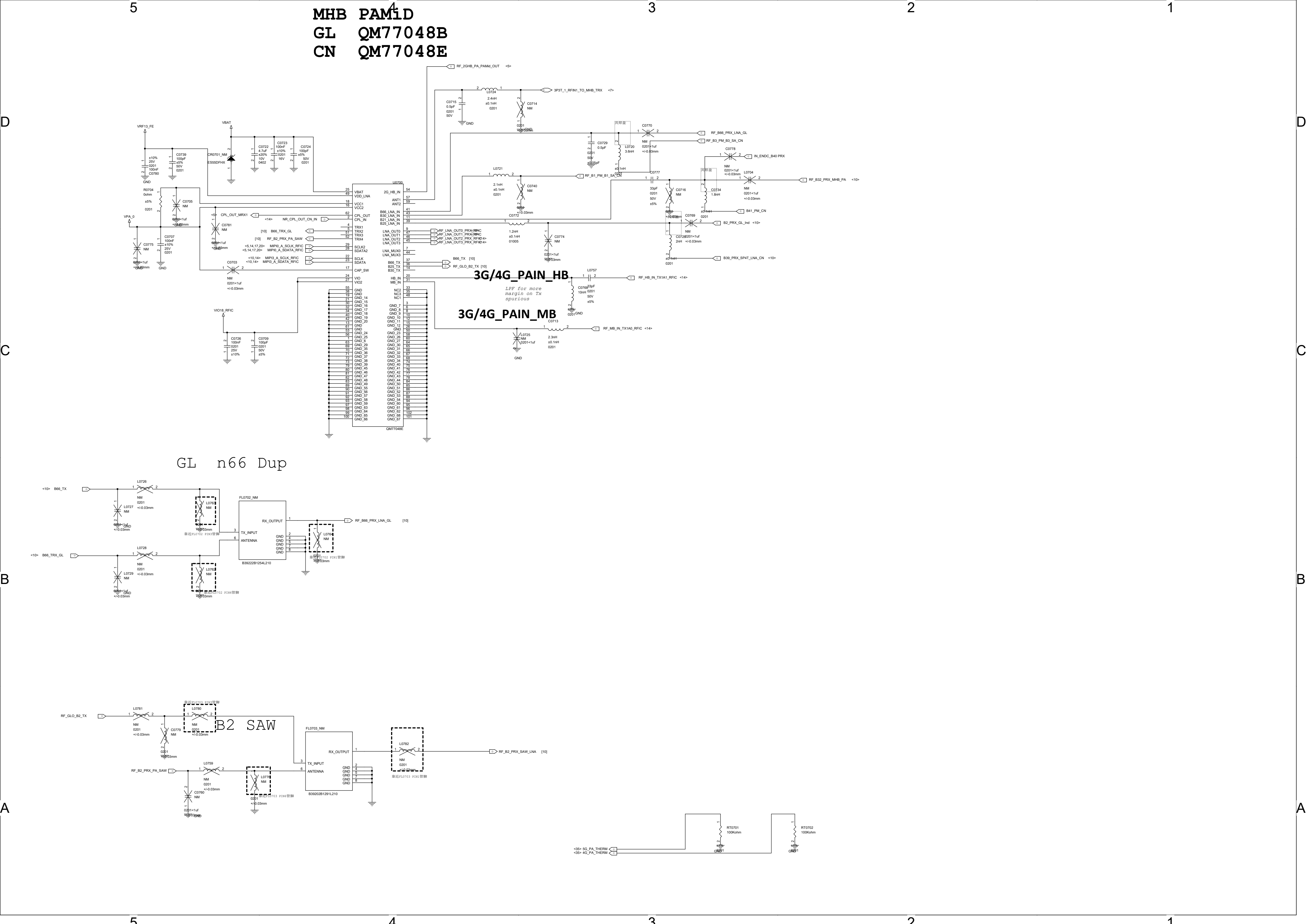
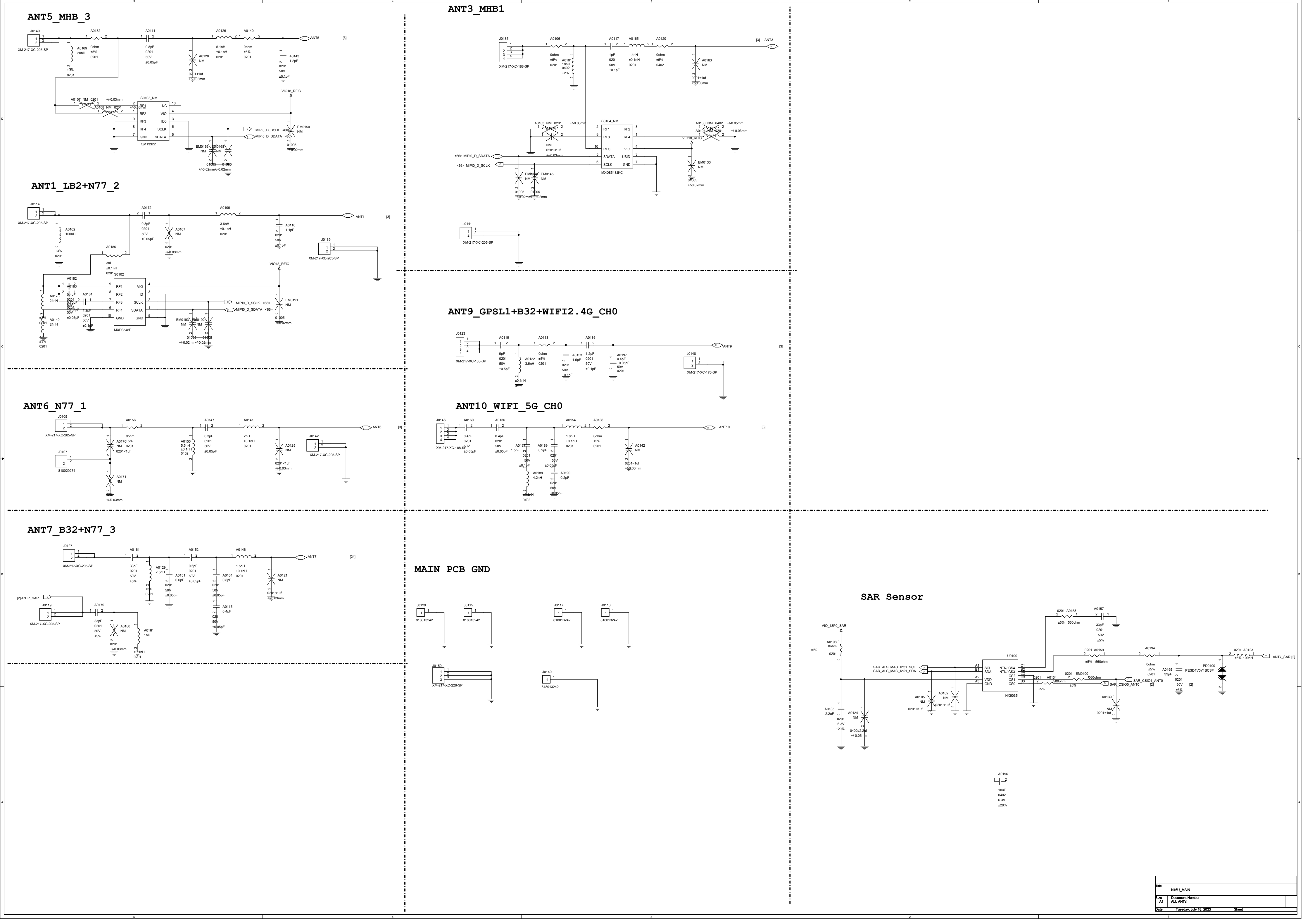
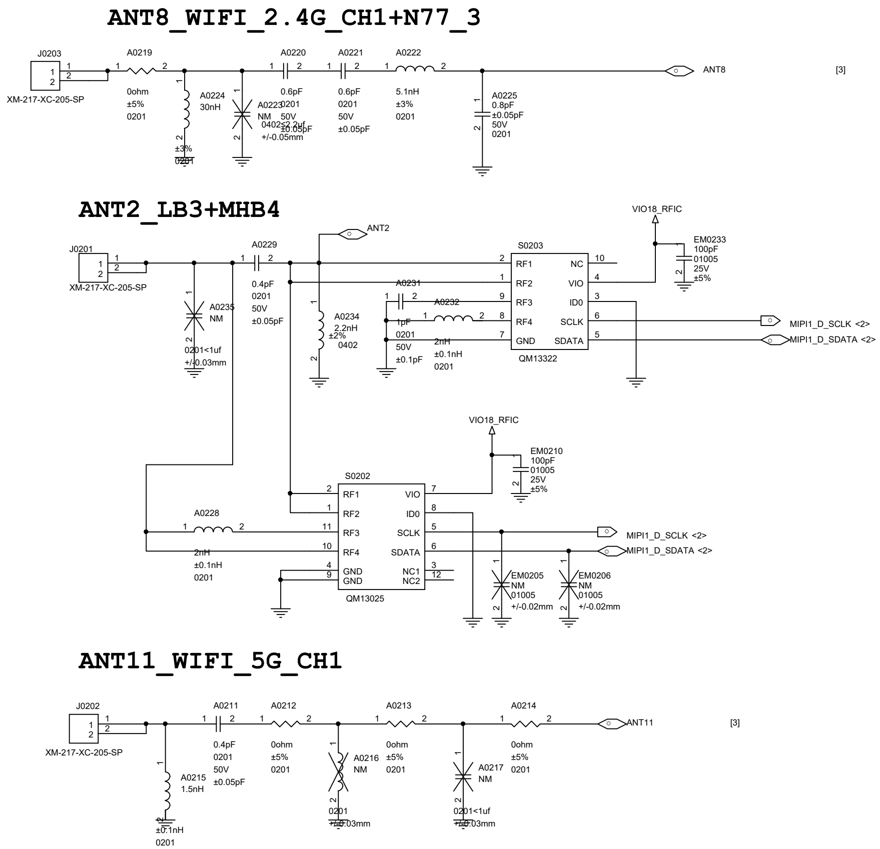


MHB PAMiD
GL QM77048B
CN QM77048E
U0700
QM77048E
VBAT
VRF13_FE
VPA_0
VIO18_RFIC
RF_2GHB_PA_PAMid_OUT
3P3T_1_RFIN1_TO_MHB_TRX
RF_B66_PRX_LNA_GL
RF_B3_PM_B3_SA_CN
IN_ENDC_B40 PRX
RF_B1_PM_B1_SA_CN
RF_B32_PRX_MHB_PA
B41_PM_CN
B2_PRX_GL_Ind
B39_PRX_SP4T_LNA_CN
CPL_OUT_MRX1
NR_CPL_OUT_CN_IN
B66_TRX_GL
RF_B2_PRX_PA_SAW
MIPI0_A_SCLK_RFIC
MIPI0_A_SDATA_RFIC
MIPI3_A_SCLK_RFIC
MIPI3_A_SDATA_RFIC
RF_LNA_OUT0_PRX_RFIC
RF_LNA_OUT1_PRX_RFIC
RF_LNA_OUT2_PRX_RFIC
RF_LNA_OUT3_PRX_RFIC
B66_TX
RF_GLO_B2_TX
3G/4G_PAIN_HB
LPF for more margin on Tx spurious
RF_HB_IN_TX1A1_RFIC
3G/4G_PAIN_MB
RF_MB_IN_TX1A0_RFIC
GL n66 Dup
FL0702_NM
B39222B1254L210
RF_B66_PRX_LNA_GL
B2 SAW
FL0703_NM
B39202B1291L210
RF_B2_PRX_SAW_LNA
5G_PA_THERM
4G_PA_THERM
RT0701
100Kohm
RT0702
100Kohm


ANT5_MHB_3
ANT3_MHB1
ANT1_LB2+N77_2
ANT9_GPSL1+B32+WIFI2.4G_CH0
ANT6_N77_1
ANT10_WIFI_5G_CH0
ANT7_B32+N77_3
MAIN PCB GND
SAR Sensor
Title
N16U_MAIN
Size
A1
Document Number
ALL ANTs'
Date: Tuesday, July 18, 2023
Sheet



Title	N16U_MAIN
Size A2	Document Number ANT2
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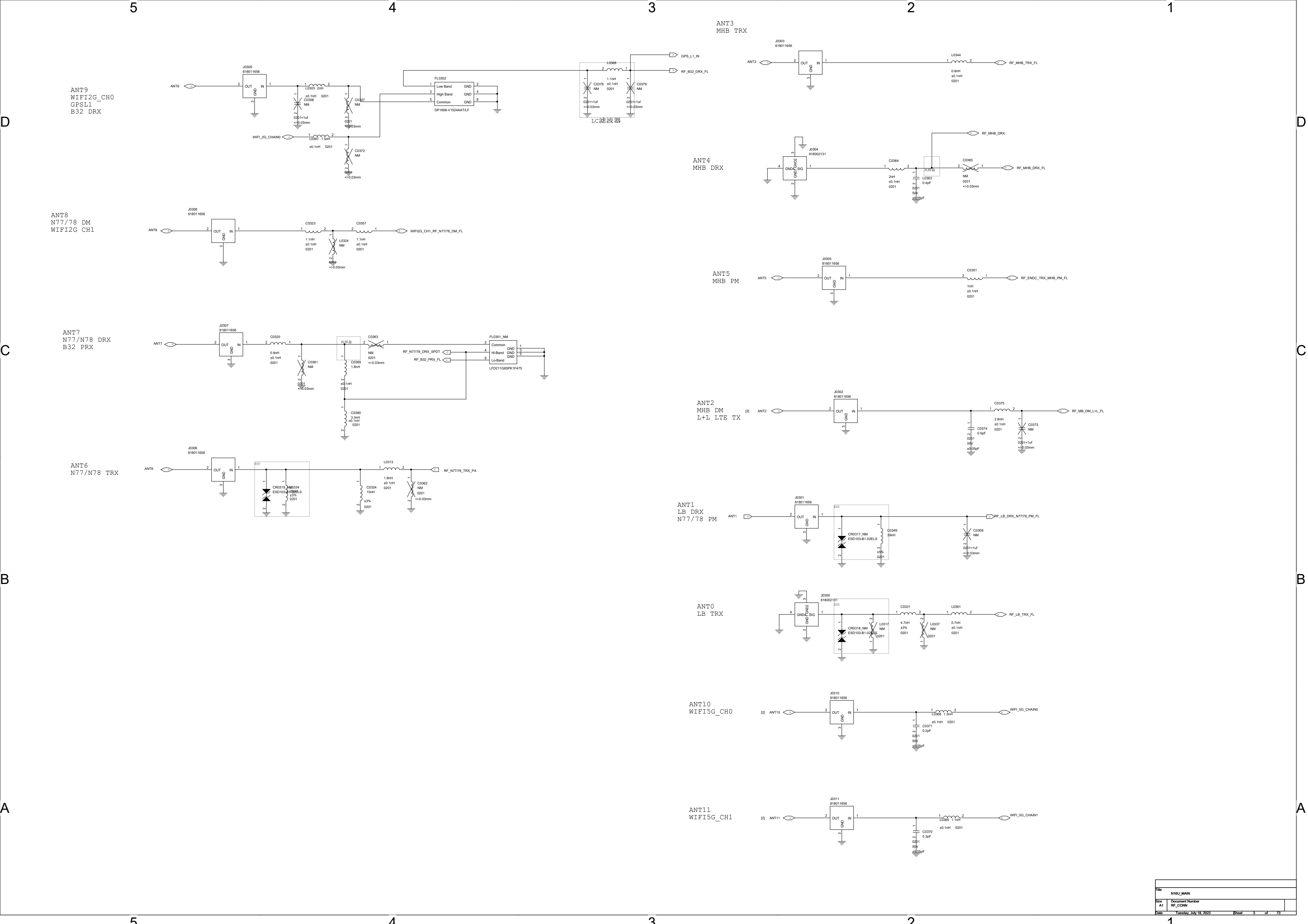
ANT9
WIFI2G_CH0
GPSL1
B32 DRX
J0309
818011656
L0303 2nH
C0306 NM
C0307 NM
FL0302
Low Band
High Band
Common
DP1608-V1524AAT/LF
L0368
C0378 NM
C0379 NM
LC滤波器
GPS_L1_IN
RF_B32_DRX_FL
WIFI_2G_CHAIN0
L0367 1.5nH
C0372 NM
ANT8
N77/78 DM
WIFI2G CH1
J0308
818011656
C0323
C0357
L0324 NM
WIFI2G_CH1_RF_N77/78_DM_FL
ANT7
N77/N78 DRX
B32 PRX
J0307
818011656
C0320
C0361 NM
C0363 NM
C0369 1.8nH
C0380 3.3nH
FL0301_NM
Hi-Band
Lo-Band
LFD211G65PK1F475
RF_N77/78_DRX_SPDT
RF_B32_PRX_FL
ANT6
N77/N78 TRX
J0306
818011656
CR0319
L0334
L0313
C0324 10nH
C0362 NM
RF_N77/78_TRX_PA
ANT3
MHB TRX
J0303
818011656
L0344
RF_MHB_TRX_FL
ANT4
MHB DRX
J0304
818002131
C0364
L0363 0.4pF
C0365 NM
RF_MHB_DRX
RF_MHB_DRX_FL
ANT5
MHB PM
J0305
818011656
C0351
RF_ENDC_TRX_MHB_PM_FL
ANT2
MHB DM
L+L LTE TX
J0302
818011656
C0375
C0374 0.5pF
C0373 NM
RF_MB_DM_L+L_FL
ANT1
LB DRX
N77/78 PM
J0301
818011656
CR0317_NM
ESD103-B1-02ELS
C0349 39nH
C0309 NM
RF_LB_DRX_N77/78_PM_FL
ANT0
LB TRX
J0300
818002131
CR0318_NM
L0317 NM
C0331 4.7nH
L0337 NM
L0361 0.7nH
RF_LB_TRX_FL
ANT10
WIFI5G_CH0
J0310
818011656
L0366 1.3nH
C0371 0.2pF
WIFI_5G_CHAIN0
ANT11
WIFI5G_CH1
J0311
818011656
L0365 1.1nH
C0370 0.3pF
WIFI_5G_CHAIN1
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N16U_MAIN
Size A1
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RF_CONN
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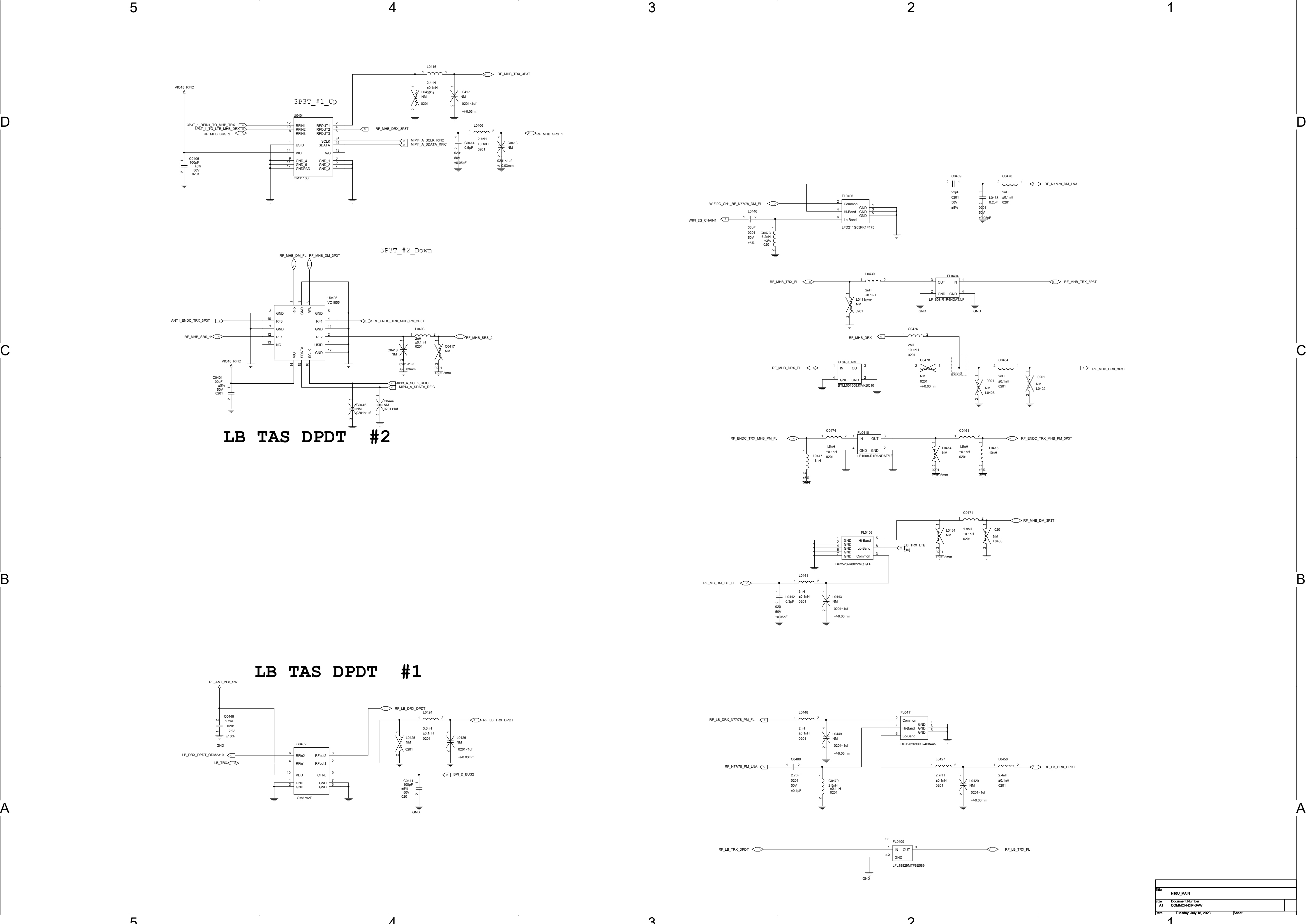
3P3T_#1_Up
U0401
QM11133
RF_MHB_TRX_3P3T
RF_MHB_DRX_3P3T
RF_MHB_SRS_1
RF_MHB_SRS_2
3P3T_1_RFIN1_TO_MHB_TRX
3P3T_1_TO_LTE_MHB_DRX
MIPI4_A_SCLK_RFIC
MIPI4_A_SDATA_RFIC
VIO18_RFIC
3P3T_#2_Down
U0403
VC1855
RF_MHB_DM_FL
RF_MHB_DM_3P3T
ANT1_ENDC_TRX_3P3T
RF_ENDC_TRX_MHB_PM_3P3T
MIPI3_A_SCLK_RFIC
MIPI3_A_SDATA_RFIC
LB TAS DPDT #2
LB TAS DPDT #1
RF_ANT_2P8_SW
S0402
OM8792F
LB_DRX_DPDT_QDM2310
LB_TRX
RF_LB_DRX_DPDT
RF_LB_TRX_DPDT
BPI_D_BUS2
FL0406
LFD211G65PK1F475
WIFI2G_CH1_RF_N77/78_DM_FL
WIFI_2G_CHAIN1
RF_N77/78_DM_LNA
FL0404
LF1608-R1R6NDAT/LF
RF_MHB_TRX_FL
RF_MHB_DRX
FL0407_NM
RF_MHB_DRX_FL
RF_MHB_DRX_3P3T
FL0410
RF_ENDC_TRX_MHB_PM_FL
RF_ENDC_TRX_MHB_PM_3P3T
FL0408
DP2520-R0822MQT/LF
RF_MHB_DM_3P3T
LB_TRX_LTE
RF_MB_DM_L+L_FL
FL0411
DPX202690DT-4084A5
RF_LB_DRX_N77/78_PM_FL
RF_N77/78_PM_LNA
RF_LB_DRX_DPDT
FL0409
LFL18829MTF8E589
RF_LB_TRX_DPDT
RF_LB_TRX_FL
Title
N16U_MAIN
Size
A1
Document Number
COMMON-DIP-SAW
Date:
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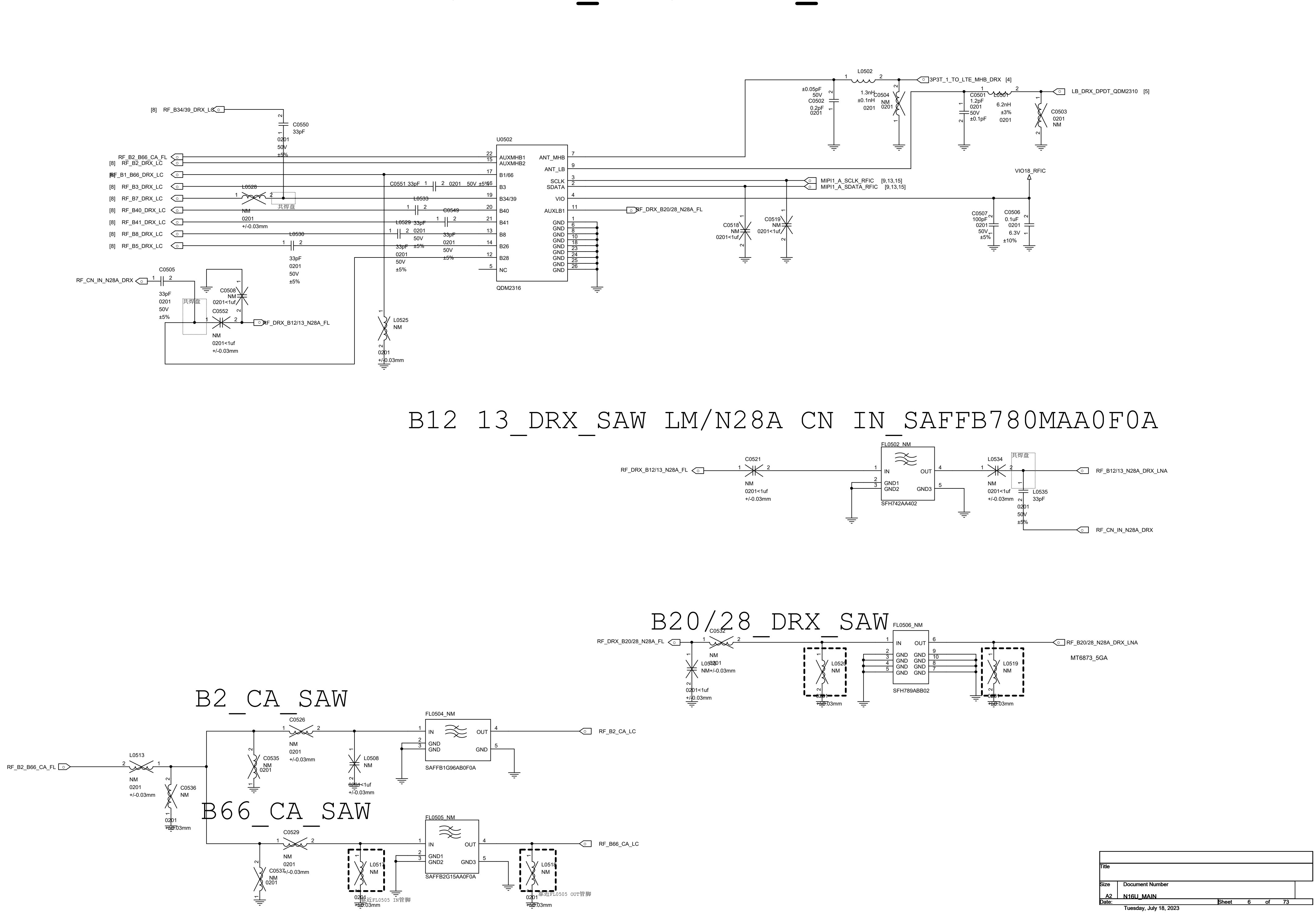
Sheet

DiFEM QDM2315_GL/QDM2316_CN



Title	
Size A2	Document Number N16U_MAIN
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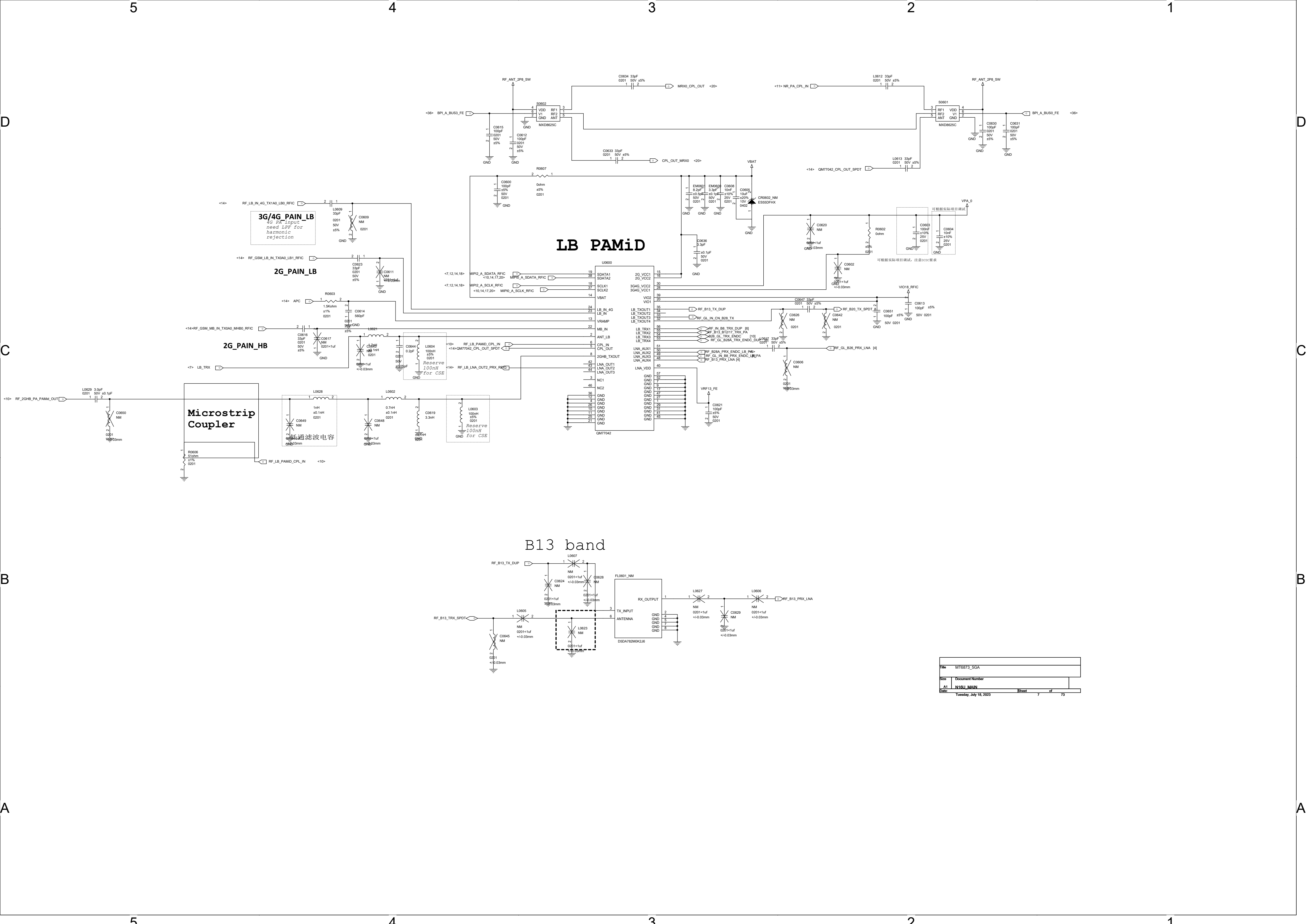

LB PAMiD
3G/4G_PAIN_LB
4G PA input need LPF for harmonic rejection
2G_PAIN_LB
2G_PAIN_HB
Microstrip Coupler
B13 band
U0600
QM77042
S0602
MXD8625C
S0601
MXD8625C
FL0601_NM
D5DA782M0K2J6
RF_ANT_2P8_SW
BPI_A_BUS3_FE
BPI_A_BUS0_FE
MRX0_CPL_OUT
CPL_OUT_MRX0
NR_PA_CPL_IN
QM77042_CPL_OUT_SPDT
VBAT
VPA_0
VIO18_RFIC
VRF13_FE
RF_LB_IN_4G_TX1A0_LB0_RFIC
RF_GSM_LB_IN_TX0A0_LB1_RFIC
RF_GSM_MB_IN_TX0A0_MHB0_RFIC
APC
LB_TRX
RF_2GHB_PA_PAMid_OUT
RF_LB_PAMID_CPL_IN
RF_LB_LNA_OUT2_PRX_RFIC
MIPI2_A_SDATA_RFIC
MIPI0_A_SDATA_RFIC
MIPI2_A_SCLK_RFIC
MIPI0_A_SCLK_RFIC
RF_B13_TX_DUP
RF_GL_IN_CN_B28_TX
RF_B20_TX_SPDT
RF_GL_B26_PRX_LNA
RF_B13_TRX_SPDT
RF_B13_PRX_LNA
Reserve 100nH for CSE
Title MT6873_5GA
Size A1
Document Number N16U_MAIN
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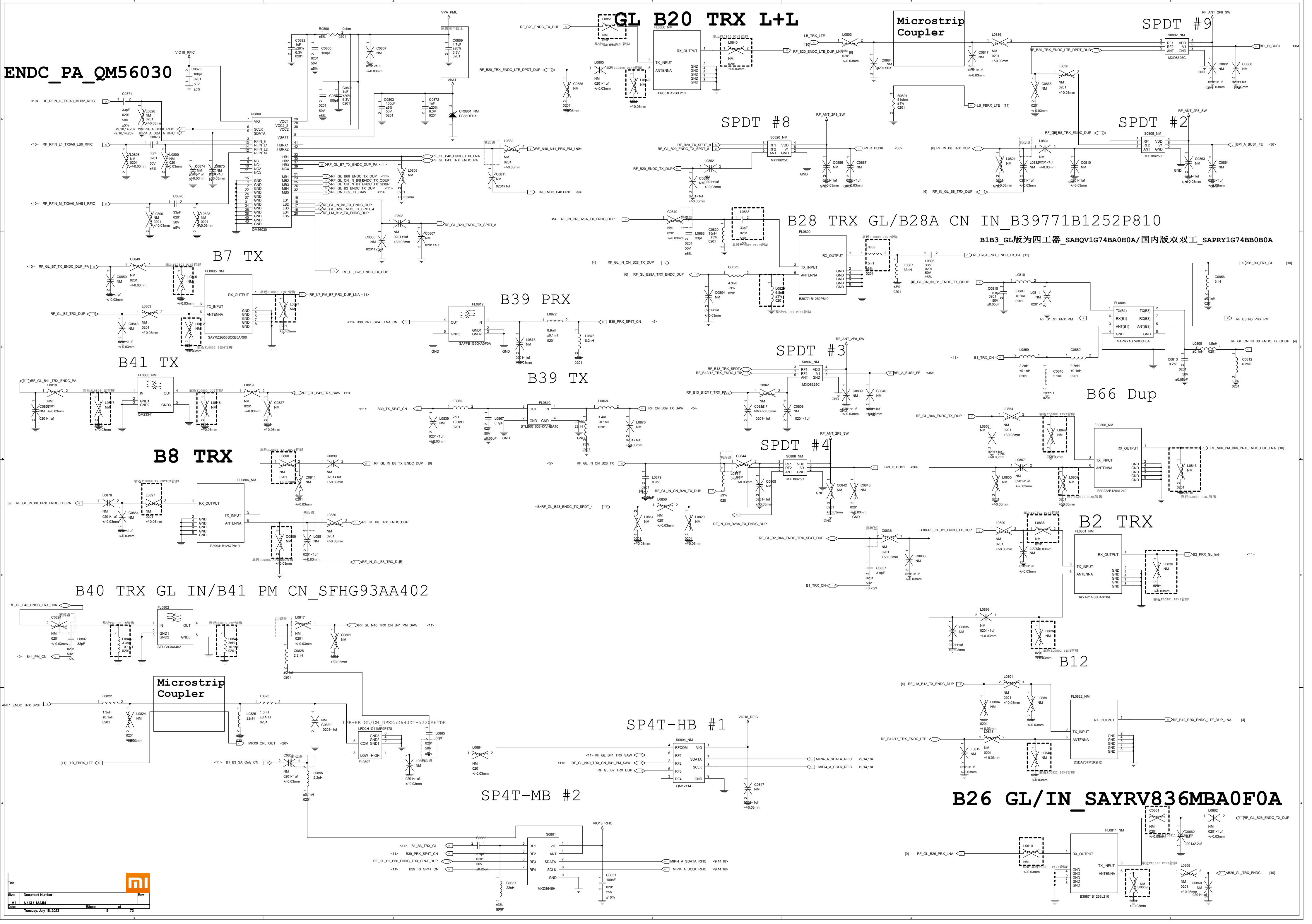

ENDC_PA_QM56030
GL B20 TRX L+L
Microstrip Coupler
SPDT #9
SPDT #8
SPDT #2
B28 TRX GL/B28A CN IN_B39771B1252P810
B1B3_GL版为四工器_SAHQV1G74BA0H0A/国内版双双工_SAPRY1G74BB0B0A
B7 TX
B39 PRX
SPDT #3
B41 TX
B39 TX
B66 Dup
B8 TRX
SPDT #4
B2 TRX
B40 TRX GL IN/B41 PM CN_SFHG93AA402
B12
Microstrip Coupler
SP4T-HB #1
SP4T-MB #2
B26 GL/IN_SAYRV836MBA0F0A
LMB+HB GL/CN_DPX252690DT-5220A6TDK
Title
Size A1
Document Number N16U_MAIN
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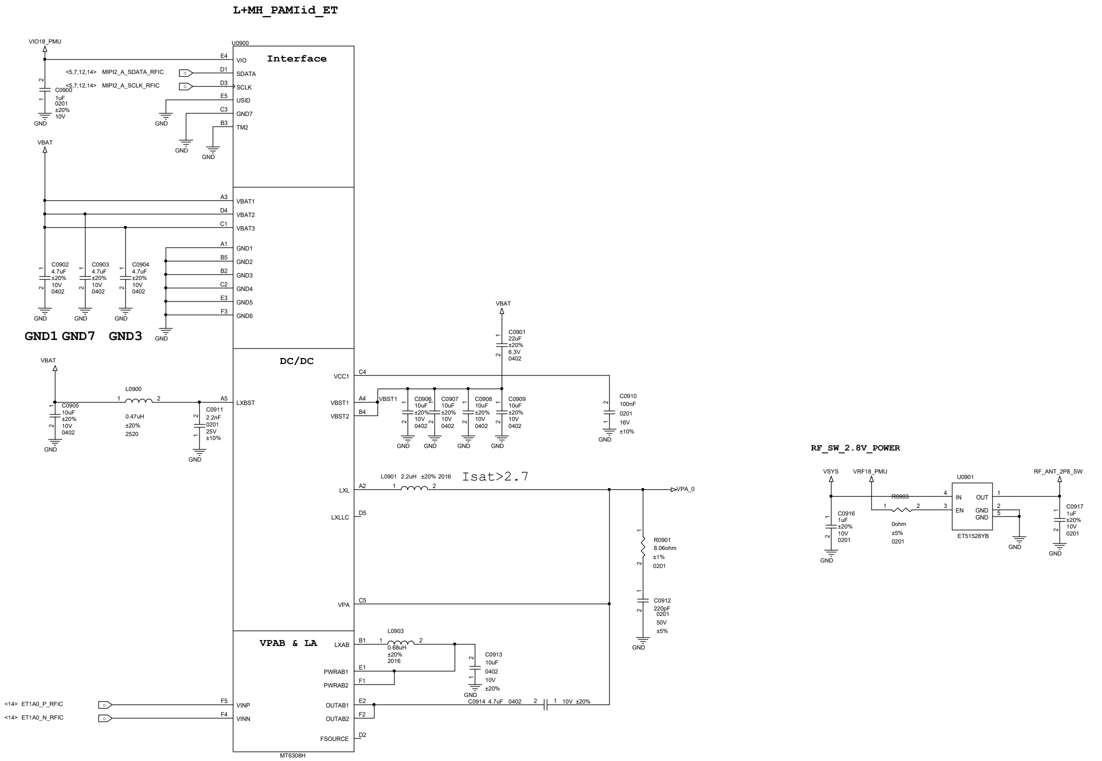

Title	N16U_MAIN	
Size A1	Document Number DCDC	
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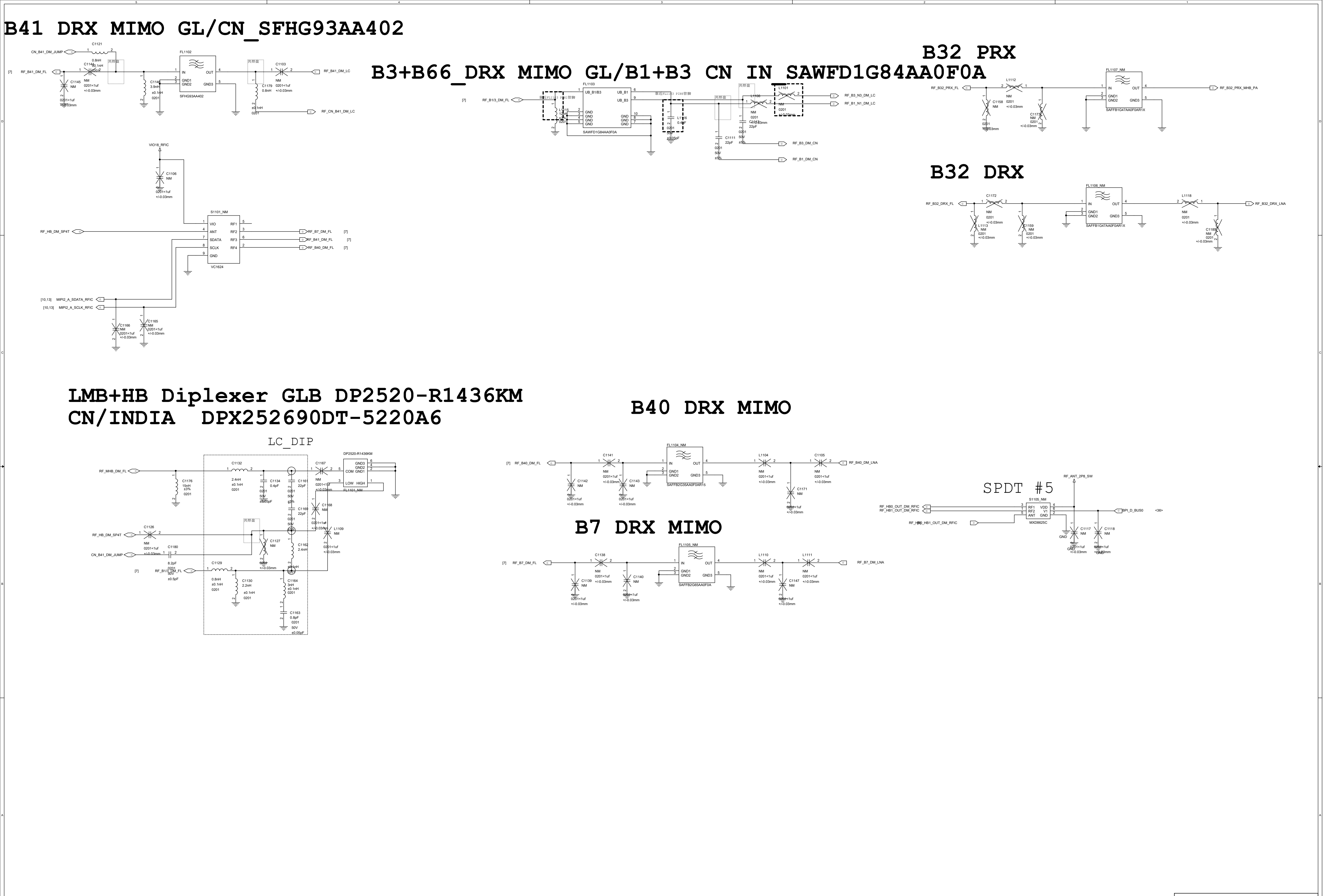

Title: N16U_MAIN
Size: A1
Document Number: DM_ASM_FILTER
Date: Tuesday, July 18, 2023
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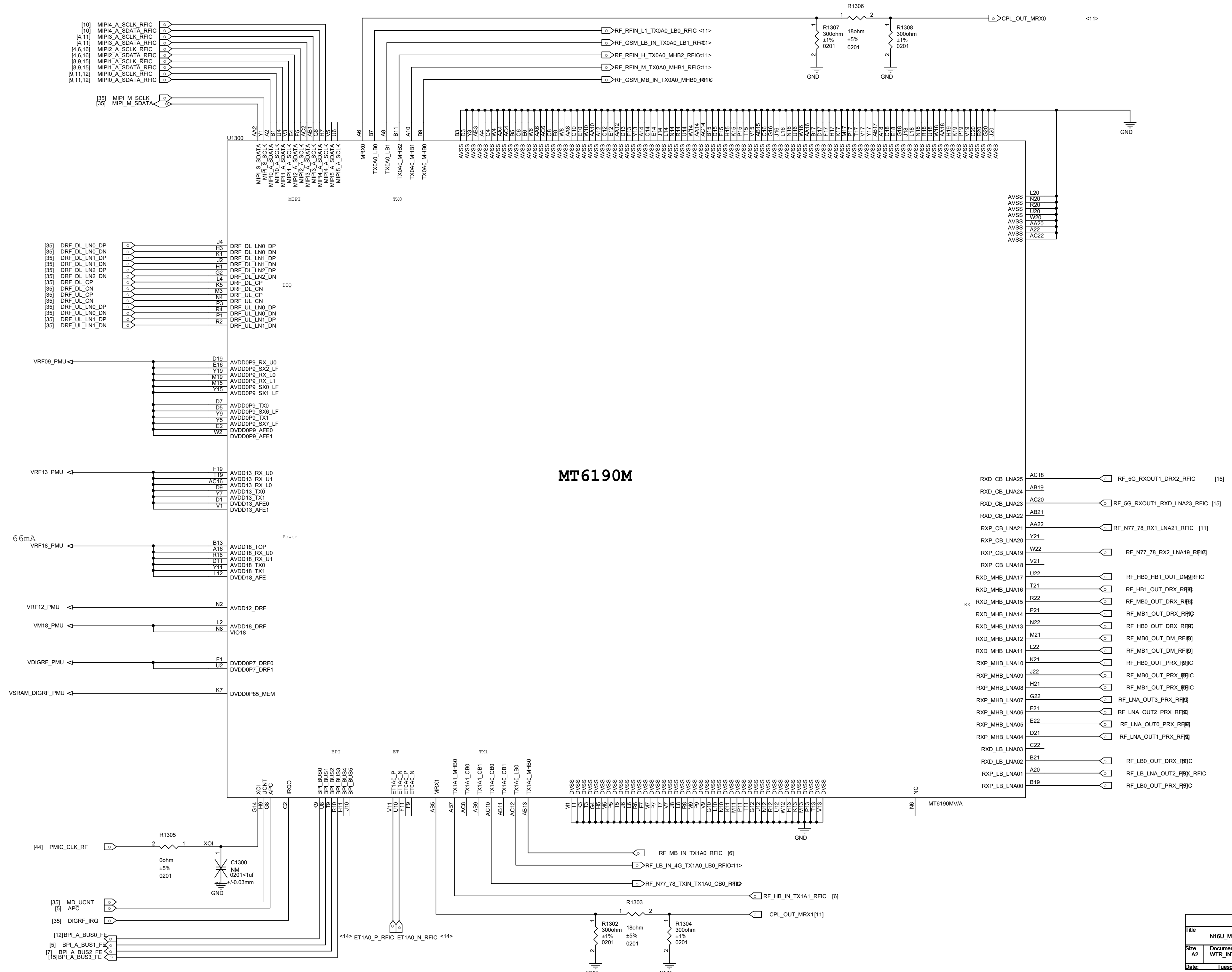

Title	N16U_MAIN
Size A2	Document Number WTR_INTERFACE
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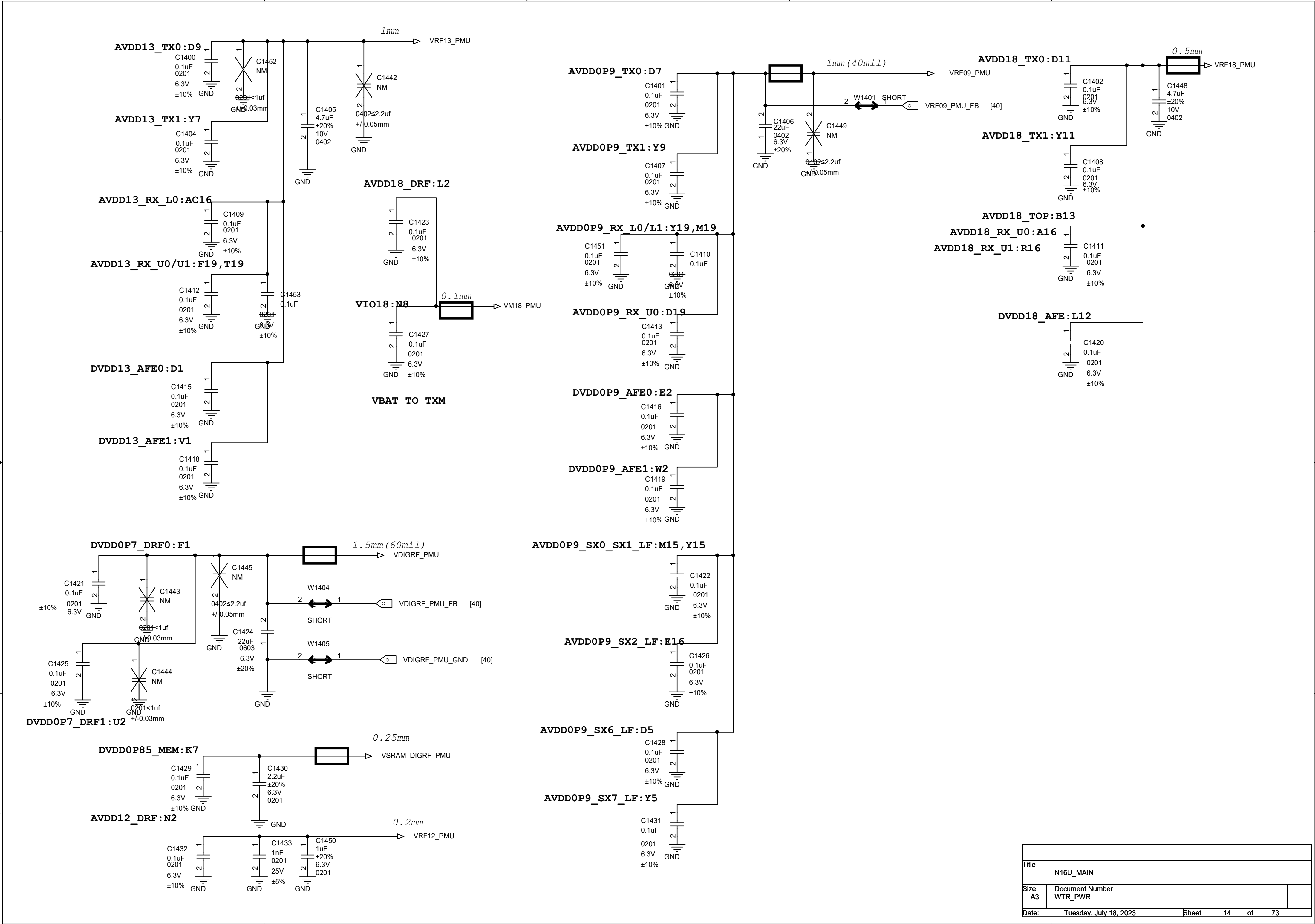

AVDD13_TX0:D9
1mm
VRF13_PMU
C1400 0.1uF 0201 6.3V ±10% GND
C1452 NM 0201<1uf +/-0.03mm GND
C1405 4.7uF ±20% 10V 0402 GND
C1442 NM 0402≤2.2uf +/-0.05mm GND
AVDD13_TX1:Y7
C1404 0.1uF 0201 6.3V ±10% GND
AVDD13_RX_L0:AC16
C1409 0.1uF 0201 6.3V ±10% GND
AVDD13_RX_U0/U1:F19,T19
C1412 0.1uF 0201 6.3V ±10% GND
C1453 0.1uF 0201 6.3V ±10% GND
DVDD13_AFE0:D1
C1415 0.1uF 0201 6.3V ±10% GND
DVDD13_AFE1:V1
C1418 0.1uF 0201 6.3V ±10% GND
AVDD18_DRF:L2
C1423 0.1uF 0201 6.3V ±10% GND
VIO18:N8
0.1mm
VM18_PMU
C1427 0.1uF 0201 6.3V ±10% GND
VBAT TO TXM
DVDD0P7_DRF0:F1
1.5mm(60mil)
VDIGRF_PMU
C1421 0.1uF 0201 6.3V ±10% GND
C1443 NM 0201<1uf +/-0.03mm GND
C1445 NM 0402≤2.2uf +/-0.05mm GND
W1404 SHORT
VDIGRF_PMU_FB [40]
C1424 22uF 0603 6.3V ±20% GND
W1405 SHORT
VDIGRF_PMU_GND [40]
C1425 0.1uF 0201 6.3V ±10% GND
C1444 NM 0201<1uf +/-0.03mm GND
DVDD0P7_DRF1:U2
DVDD0P85_MEM:K7
0.25mm
VSRAM_DIGRF_PMU
C1429 0.1uF 0201 6.3V ±10% GND
C1430 2.2uF ±20% 6.3V 0201 GND
AVDD12_DRF:N2
0.2mm
VRF12_PMU
C1432 0.1uF 0201 6.3V ±10% GND
C1433 1nF 0201 25V ±5% GND
C1450 1uF ±20% 6.3V 0201 GND
AVDD0P9_TX0:D7
1mm(40mil)
VRF09_PMU
C1401 0.1uF 0201 6.3V ±10% GND
C1406 22uF 0402 6.3V ±20% GND
C1449 NM 0402≤2.2uf +/-0.05mm GND
W1401 SHORT
VRF09_PMU_FB [40]
AVDD0P9_TX1:Y9
C1407 0.1uF 0201 6.3V ±10% GND
AVDD0P9_RX_L0/L1:Y19,M19
C1451 0.1uF 0201 6.3V ±10% GND
C1410 0.1uF 0201 6.3V ±10% GND
AVDD0P9_RX_U0:D19
C1413 0.1uF 0201 6.3V ±10% GND
DVDD0P9_AFE0:E2
C1416 0.1uF 0201 6.3V ±10% GND
DVDD0P9_AFE1:W2
C1419 0.1uF 0201 6.3V ±10% GND
AVDD0P9_SX0_SX1_LF:M15,Y15
C1422 0.1uF 0201 6.3V ±10% GND
AVDD0P9_SX2_LF:E16
C1426 0.1uF 0201 6.3V ±10% GND
AVDD0P9_SX6_LF:D5
C1428 0.1uF 0201 6.3V ±10% GND
AVDD0P9_SX7_LF:Y5
C1431 0.1uF 0201 6.3V ±10% GND
AVDD18_TX0:D11
0.5mm
VRF18_PMU
C1402 0.1uF 0201 6.3V ±10% GND
C1448 4.7uF ±20% 10V 0402 GND
AVDD18_TX1:Y11
C1408 0.1uF 0201 6.3V ±10% GND
AVDD18_TOP:B13
AVDD18_RX_U0:A16
AVDD18_RX_U1:R16
C1411 0.1uF 0201 6.3V ±10% GND
DVDD18_AFE:L12
C1420 0.1uF 0201 6.3V ±10% GND
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Size A3
Document Number WTR_PWR
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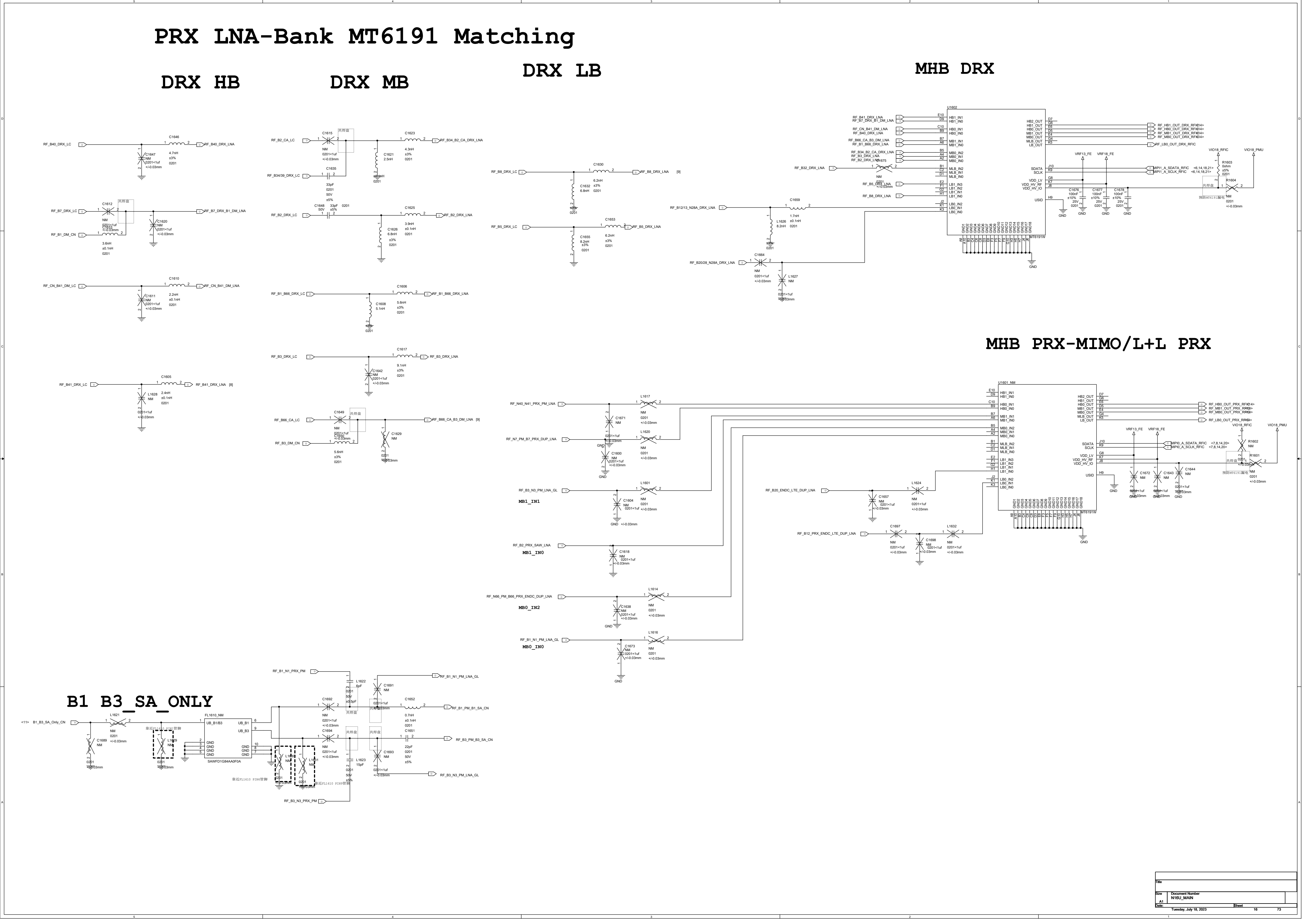
PRX LNA-Bank MT6191 Matching
DRX HB
DRX MB
DRX LB
MHB DRX
MHB PRX-MIMO/L+L PRX
B1 B3_SA_ONLY
RF_B40_DRX_LC
RF_B40_DRX_LNA
RF_B7_DRX_LC
RF_B7_DRX_B1_DM_LNA
RF_B1_DM_CN
RF_CN_B41_DM_LC
RF_CN_B41_DM_LNA
RF_B41_DRX_LC
RF_B41_DRX_LNA
RF_B2_CA_LC
RF_B34_B2_CA_DRX_LNA
RF_B34/39_DRX_LC
RF_B2_DRX_LC
RF_B2_DRX_LNA
RF_B1_B66_DRX_LC
RF_B1_B66_DRX_LNA
RF_B3_DRX_LC
RF_B3_DRX_LNA
RF_B66_CA_LC
RF_B66_CA_B3_DM_LNA
RF_B3_DM_CN
RF_B8_DRX_LC
RF_B8_DRX_LNA
RF_B5_DRX_LC
RF_B5_DRX_LNA
RF_B12/13_N28A_DRX_LNA
RF_B20/28_N28A_DRX_LNA
RF_B32_DRX_LNA
U1602
MT6191W
U1601_NM
RF_N40_N41_PRX_PM_LNA
RF_N7_PM_B7_PRX_DUP_LNA
RF_B3_N3_PM_LNA_GL
MB1_IN1
RF_B2_PRX_SAW_LNA
MB1_IN0
RF_N66_PM_B66_PRX_ENDC_DUP_LNA
MB0_IN2
RF_B1_N1_PM_LNA_GL
MB0_IN0
RF_B20_ENDC_LTE_DUP_LNA
RF_B12_PRX_ENDC_LTE_DUP_LNA
B1_B3_SA_Only_CN
FL1610_NM
SAWFD1G84AA0F0A
RF_B1_N1_PRX_PM
RF_B1_PM_B1_SA_CN
RF_B3_PM_B3_SA_CN
RF_B3_N3_PM_LNA_GL
RF_B3_N3_PRX_PM
MIPI1_A_SDATA_RFIC
MIPI1_A_SCLK_RFIC
MIPI0_A_SDATA_RFIC
MIPI0_A_SCLK_RFIC
VRF13_FE
VRF18_FE
VIO18_RFIC
VIO18_PMU
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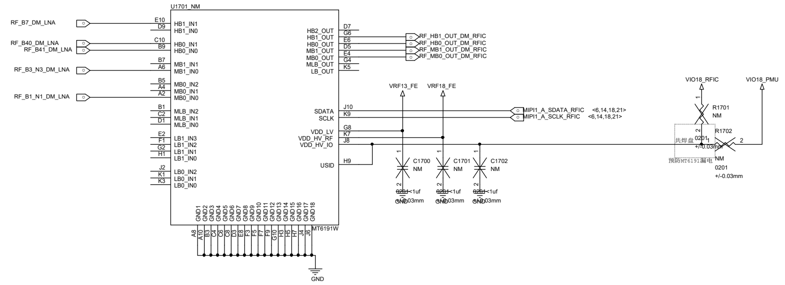
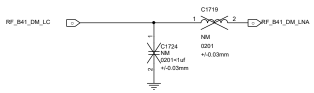
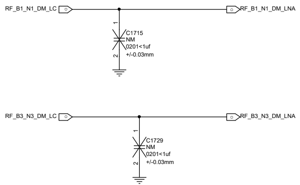
MHB DRX-MIMO



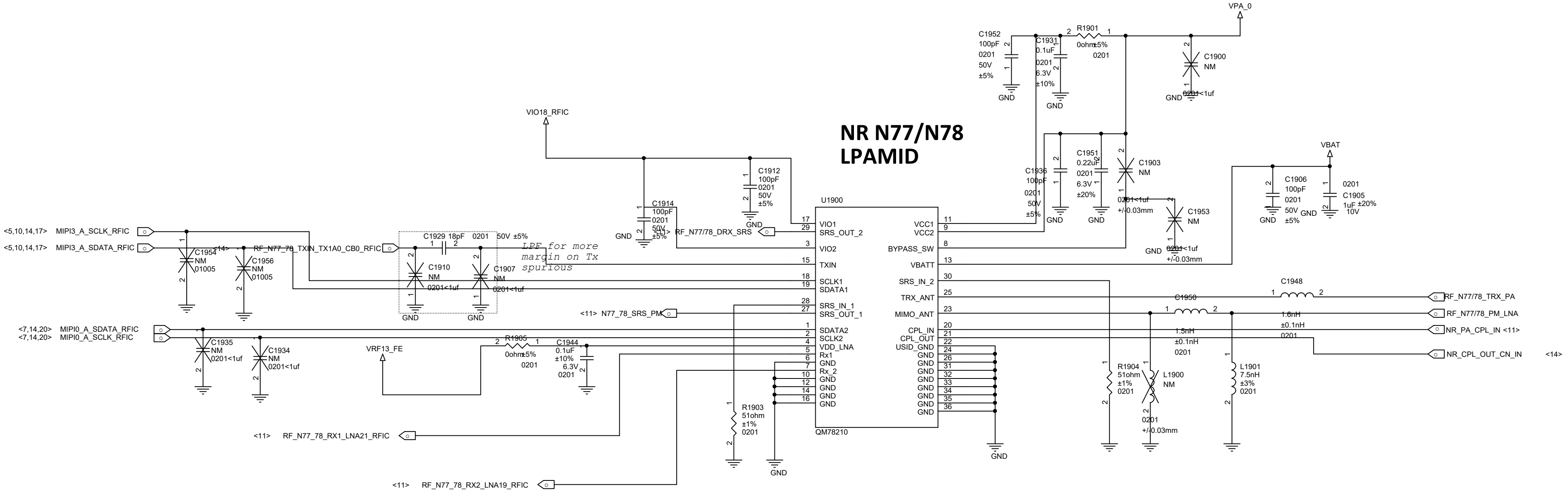
DRX-MIMO HB



DRX-MIMO MB



Title	N16U_MAIN
Size A1	Document Number LNA_DRX
Date: Tuesday, July 18, 2023	Sheet 17 73



Title	N16U_MAIN
Size A2	Document Number 5G_N77_N79_PA_PRX
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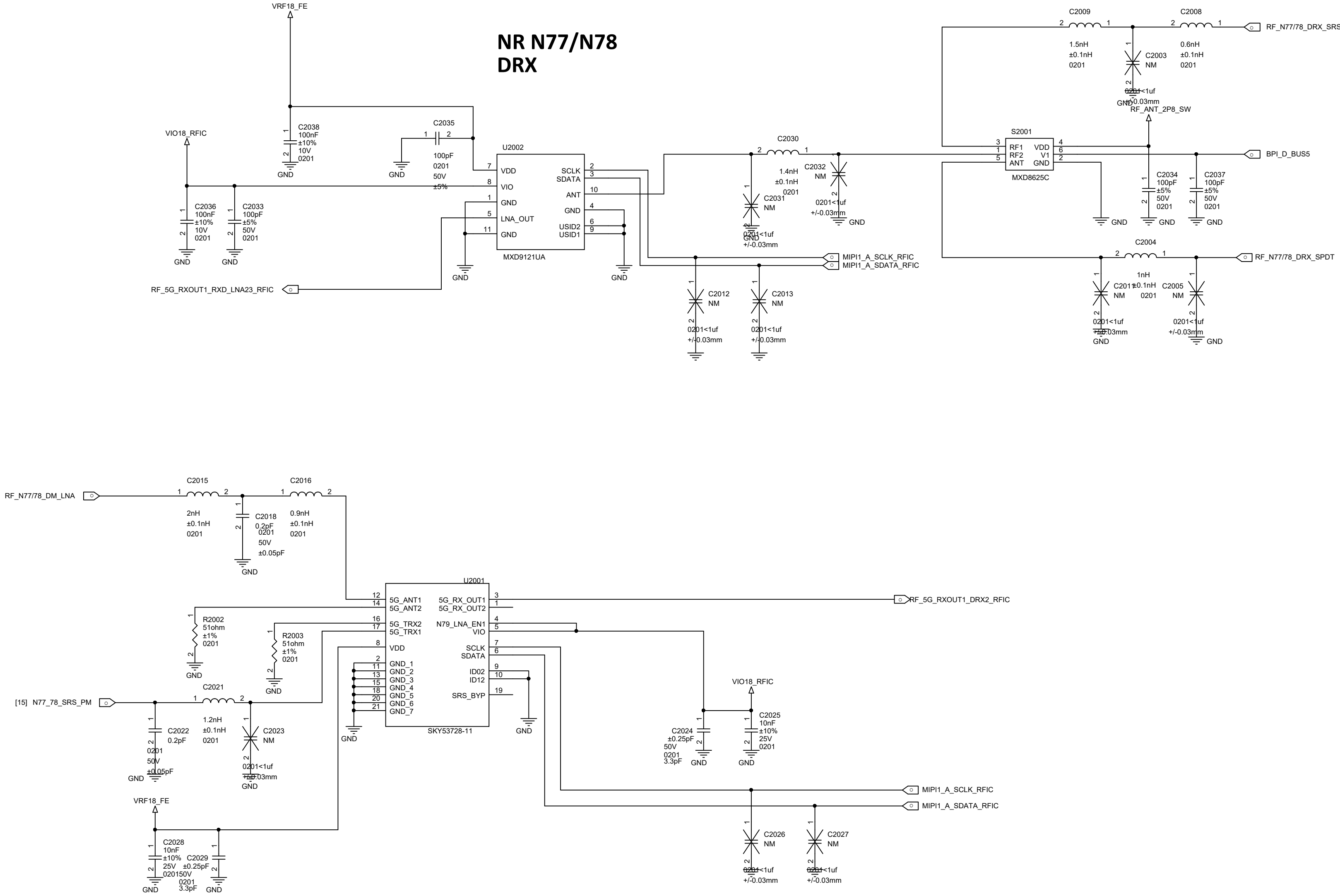
NR N77/N78 DRX



Title	N16U_MAIN
Size A2	Document Number 5G_N77_DRX_FBRX
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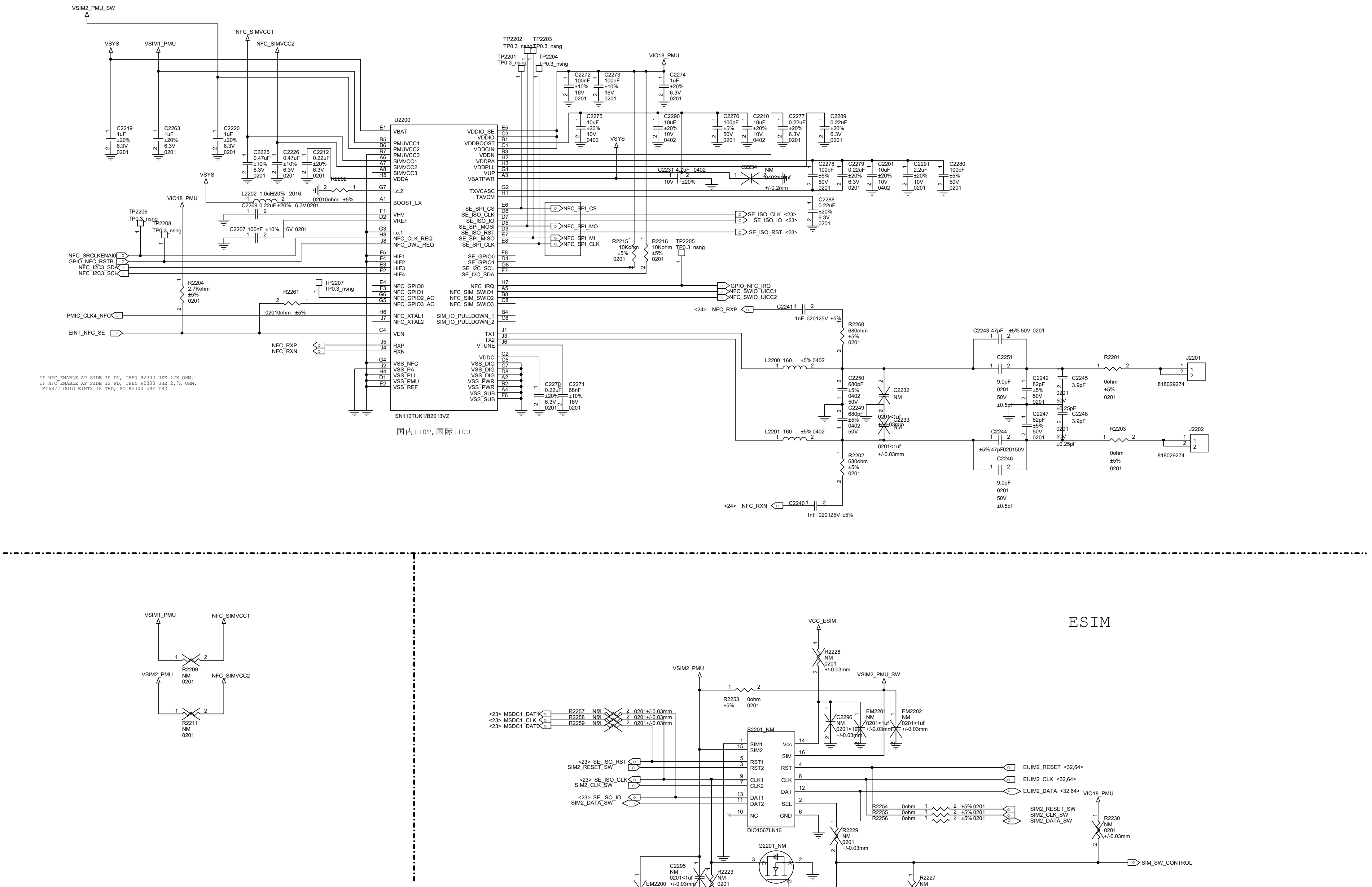

Title	N16U_MAIN
Size A2	Document Number NFC
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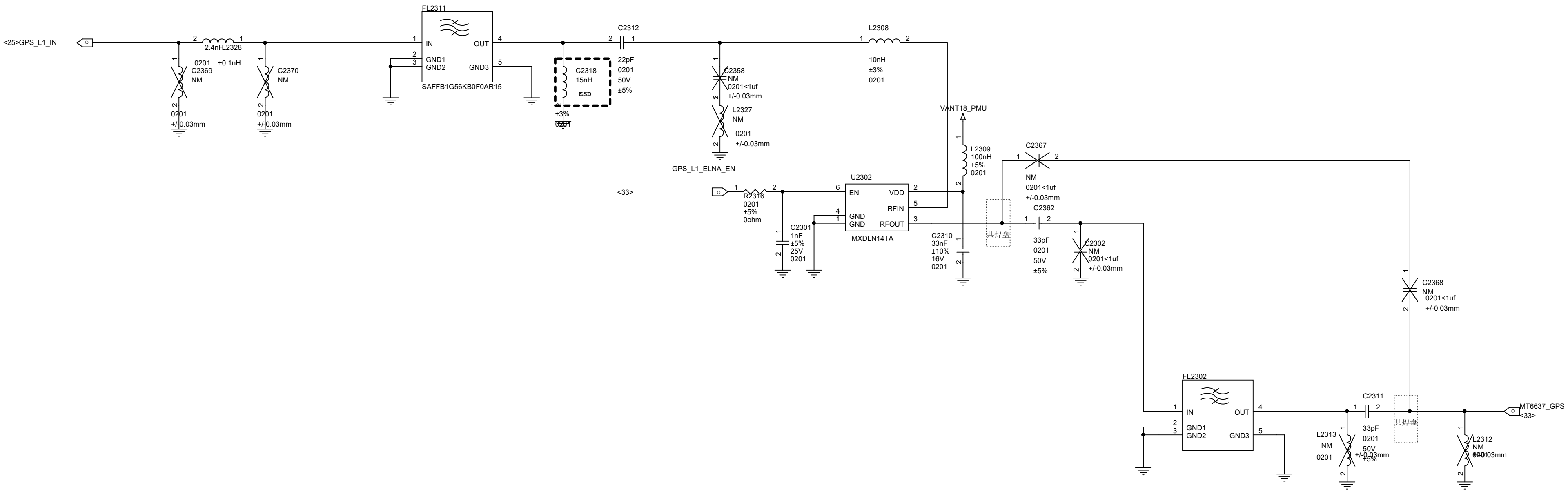

Title	N16U_MAIN
Size A2	Document Number GPS
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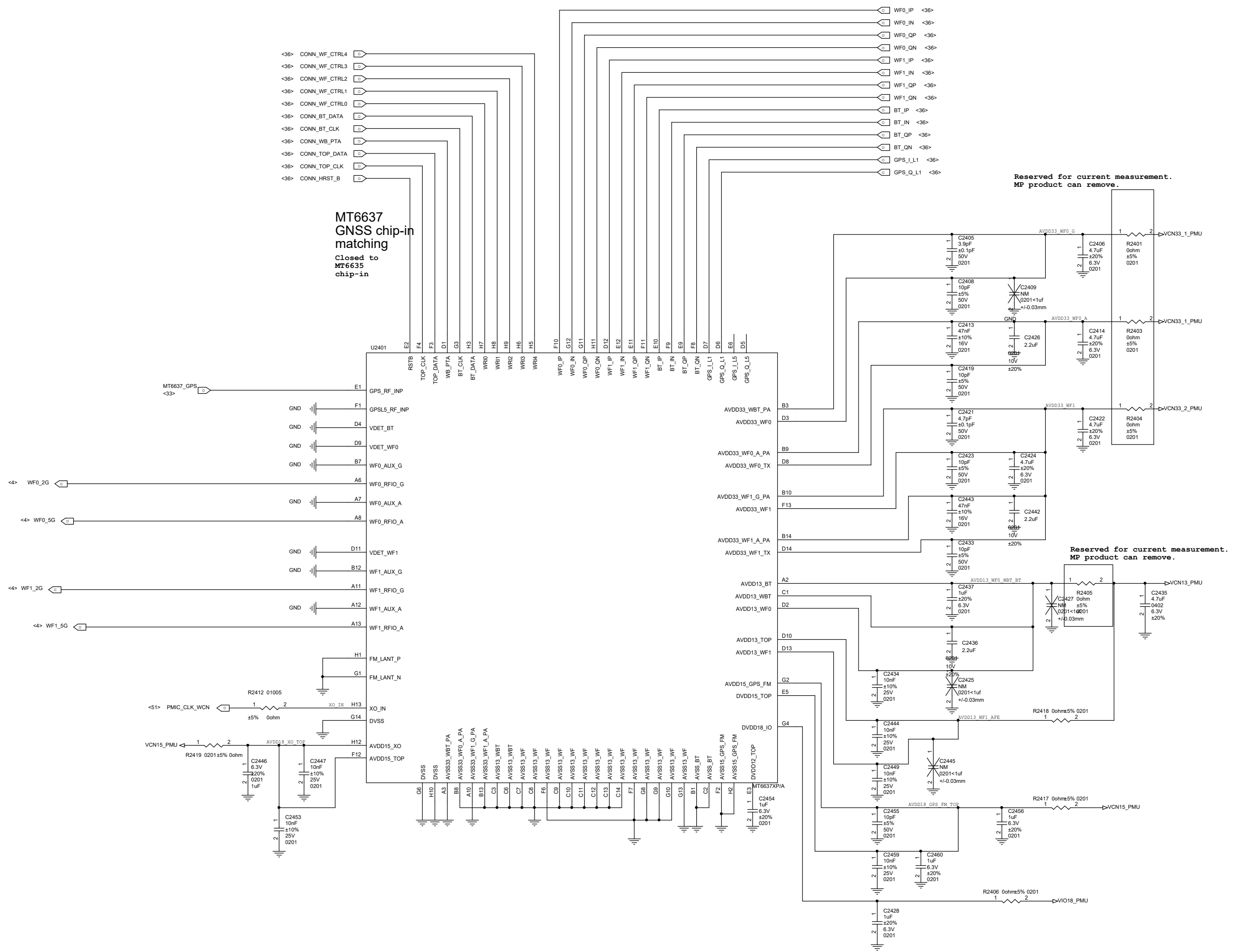

Title	N16U_MAIN
Size A1	Document Number WIFI_Interface
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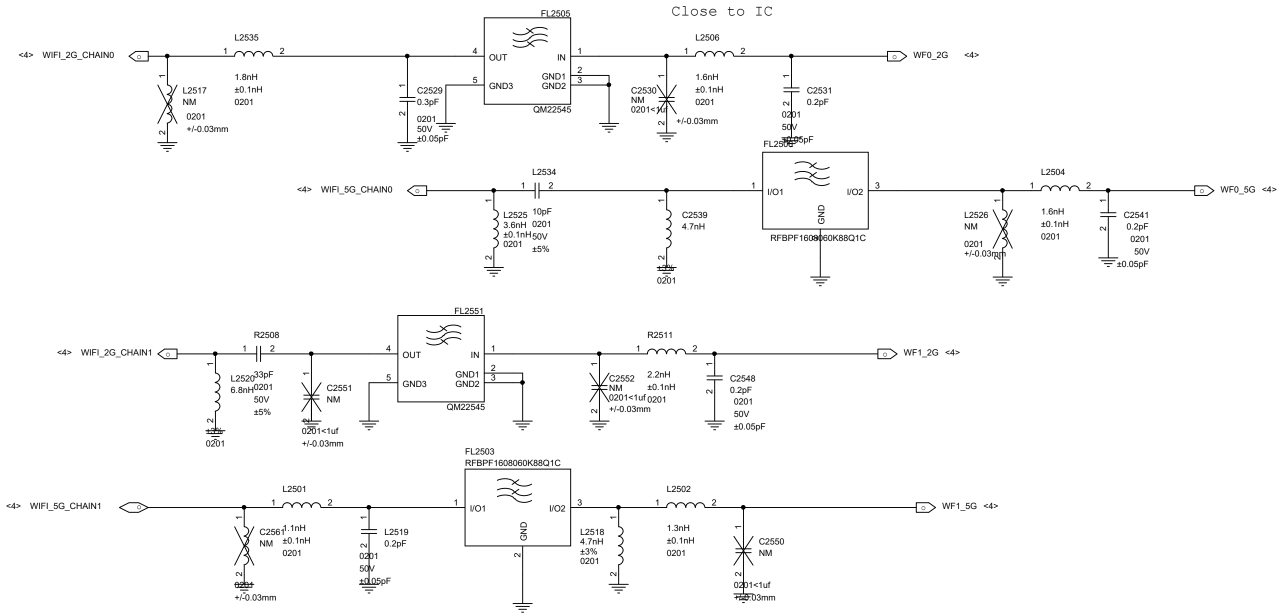
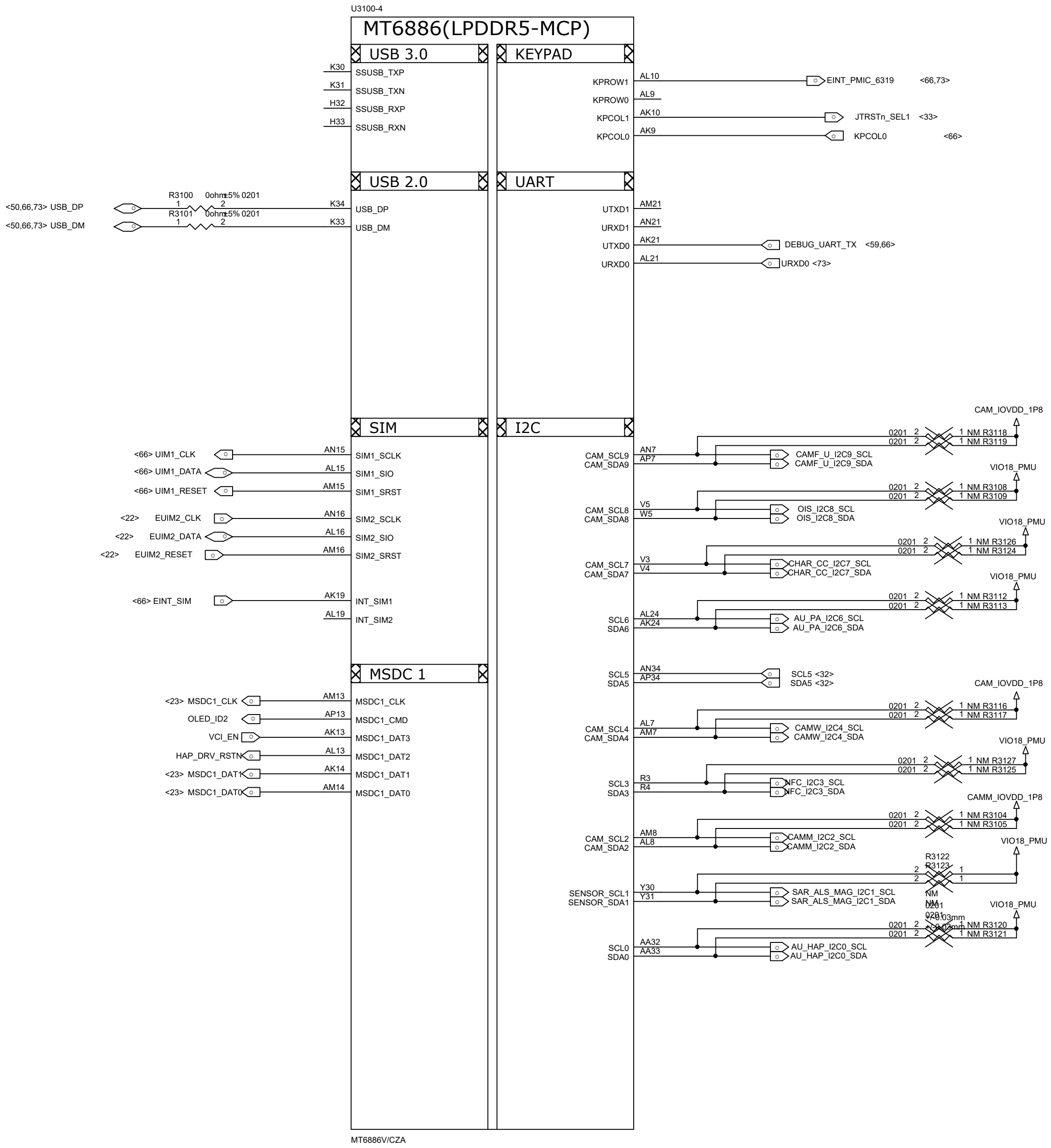
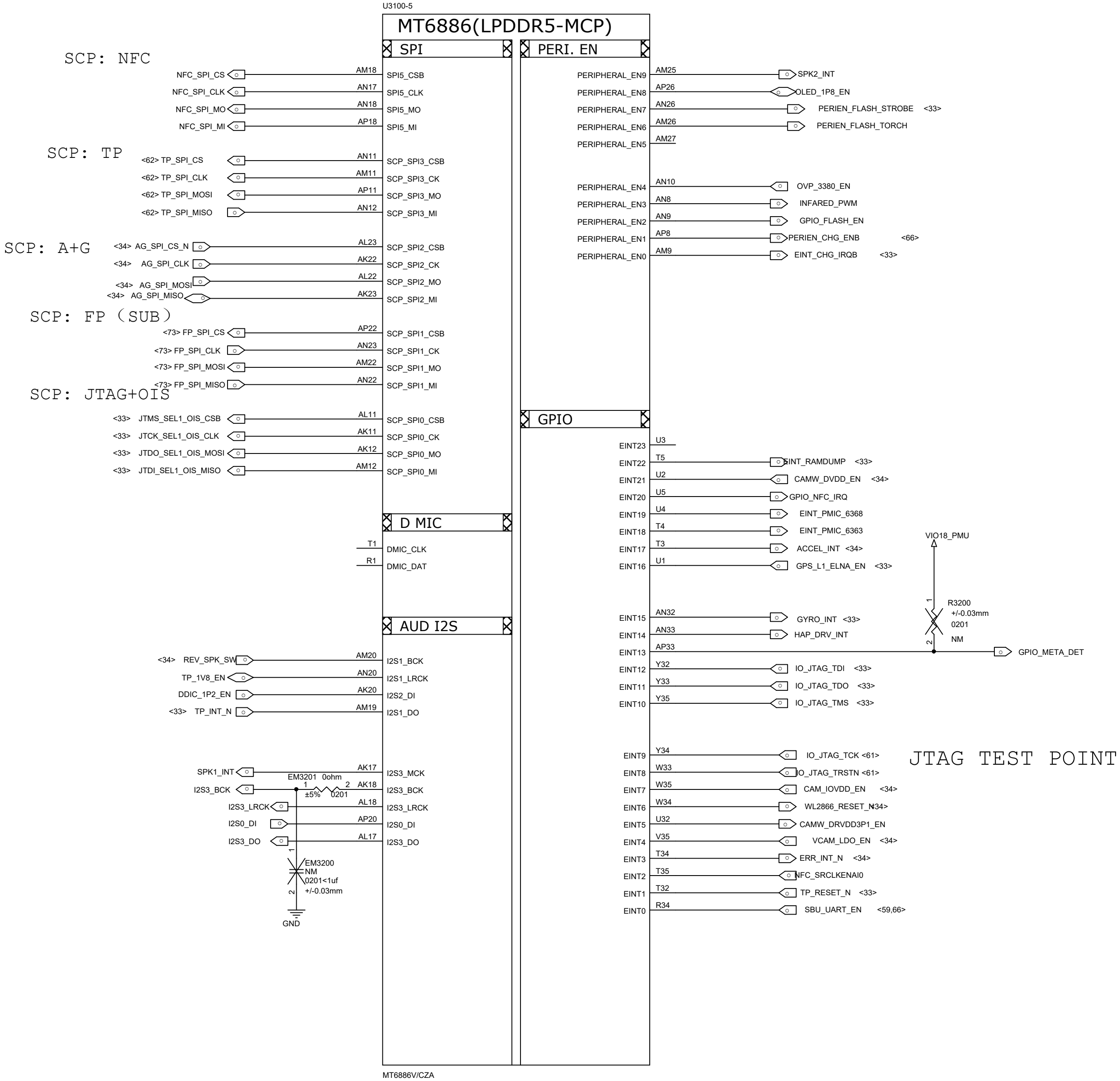


Title	N16U_MAIN
Size A2	Document Number WIFI FEM
Date: Tuesday, July 18, 2023	Sheet 25 73



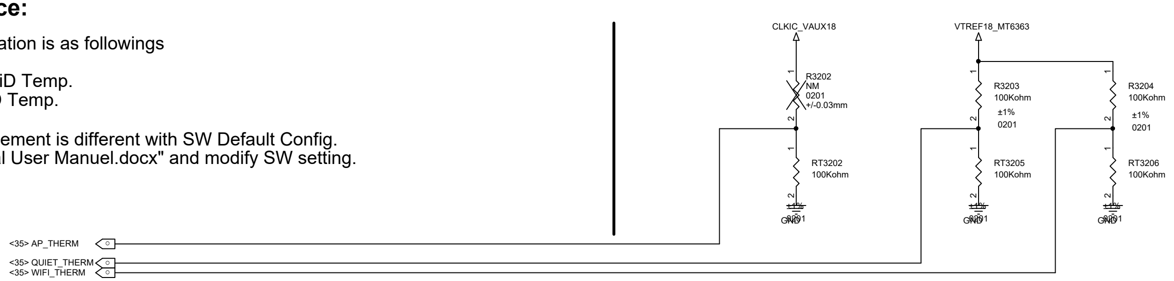
Title	N16U_MAIN		
Size A2	Document Number MT6886_GPIO_1		
Date: Tuesday, July 18, 2023	Sheet	31	73



Schematic design notice:

Note 16-1: SW Default Configuration is as followings
AUX_IN2_NTC is for AP Temp.
AUX_IN3_NTC is for MHB PAMiD Temp.
AUX_IN4_NTC is for NR PAMiD Temp.

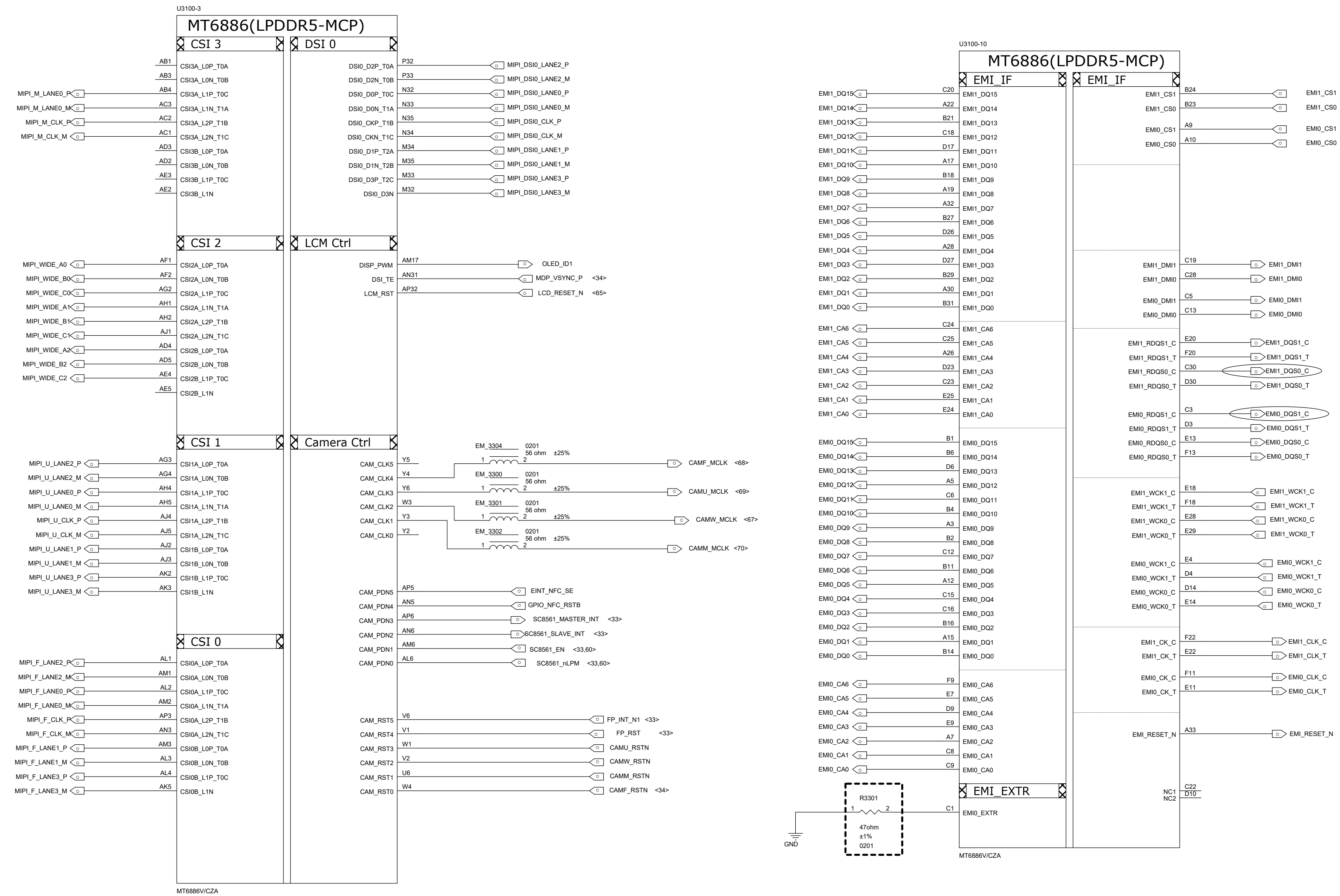
Note 16-2: If your design or placement is different with SW Default Config.
Please refer to "MTxxxx Thermal User Manuel.docx" and modify SW setting.



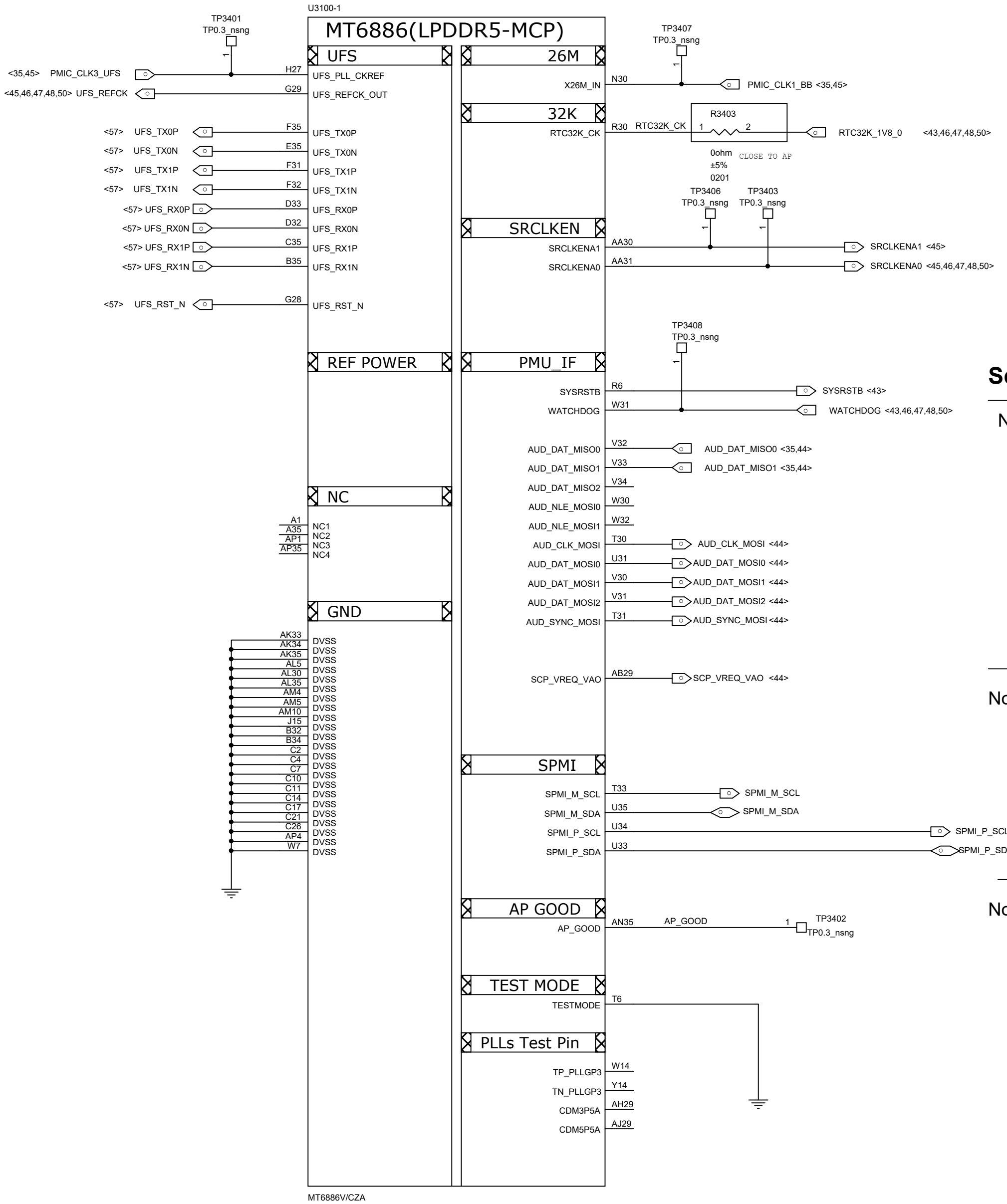
Thermistor to sense AP temperature

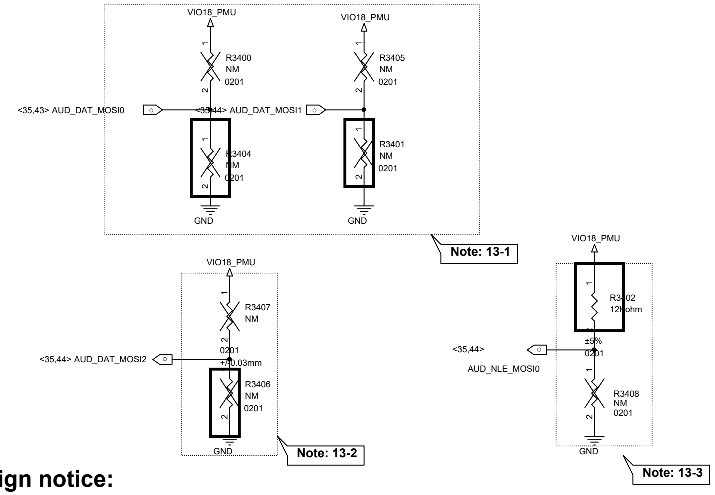
1. NTC1603 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

Title	N16U_MAIN	
Size A2	Document Number MT6886_GPIO_2	
Date:	Tuesday, July 18, 2023	Sheet 32 73



Title	N16U_MAIN
Size C	Document Number MT6886_CAM_LCM
Date: Tuesday, July 18, 2023	Sheet 33 73





Schematic design notice:

Note 13-1: "AUD_DAT_MOSI0" and "AUD_DAT_MOSI1"I pin features in trapping pin to enable JTAG.

AUD_DAT_MOSI0	AUD_DAT_MOSI1	AP JTAG	IO JTAG
L (Default)	L (Default)	N/A	N/A
L	H (by external PU)	SCP_SPI0_CSB, SCP_SPI0_CLK, SCP_SPI0_MO, SCP_SPI0_MI, KPCOL1	N/A
H (by external PU)	L	SCP_SPI0_CSB, SCP_SPI0_CLK, SCP_SPI0_MO, SCP_SPI0_MI, KPCOL1	EINT8, EINT9 EINT10, EINT11 EINT12
H (by external PU)	H (by external PU)	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

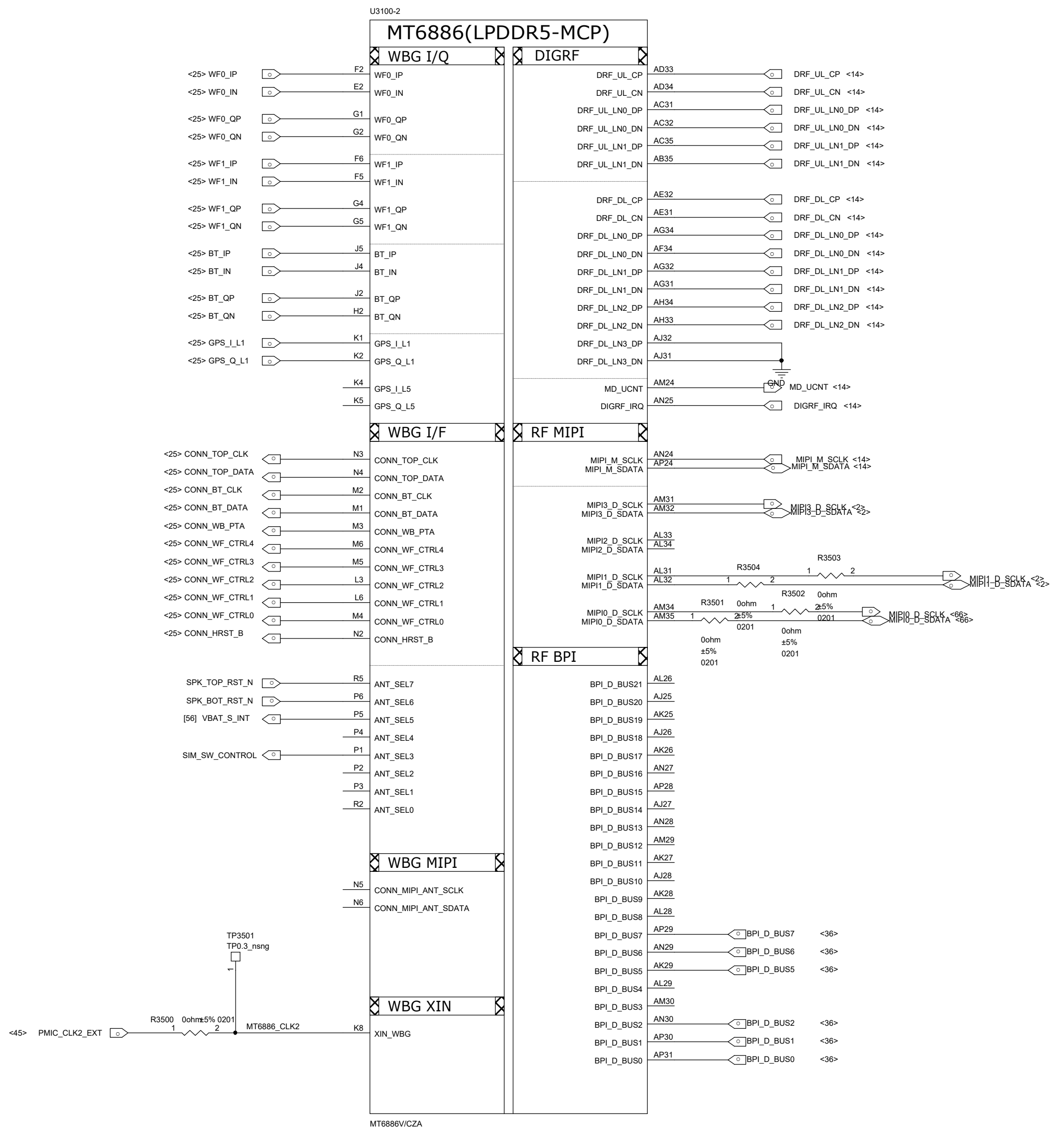
Note 13-2: "AUD_DAT_MOSI2" pins are DDR strapping for SW co-load

AUD_DAT_MOSI2	MODE
L (Default)	LP5-MCP-6400Mbps
H (by external PU)	LP4X-MCP-4266Mbps

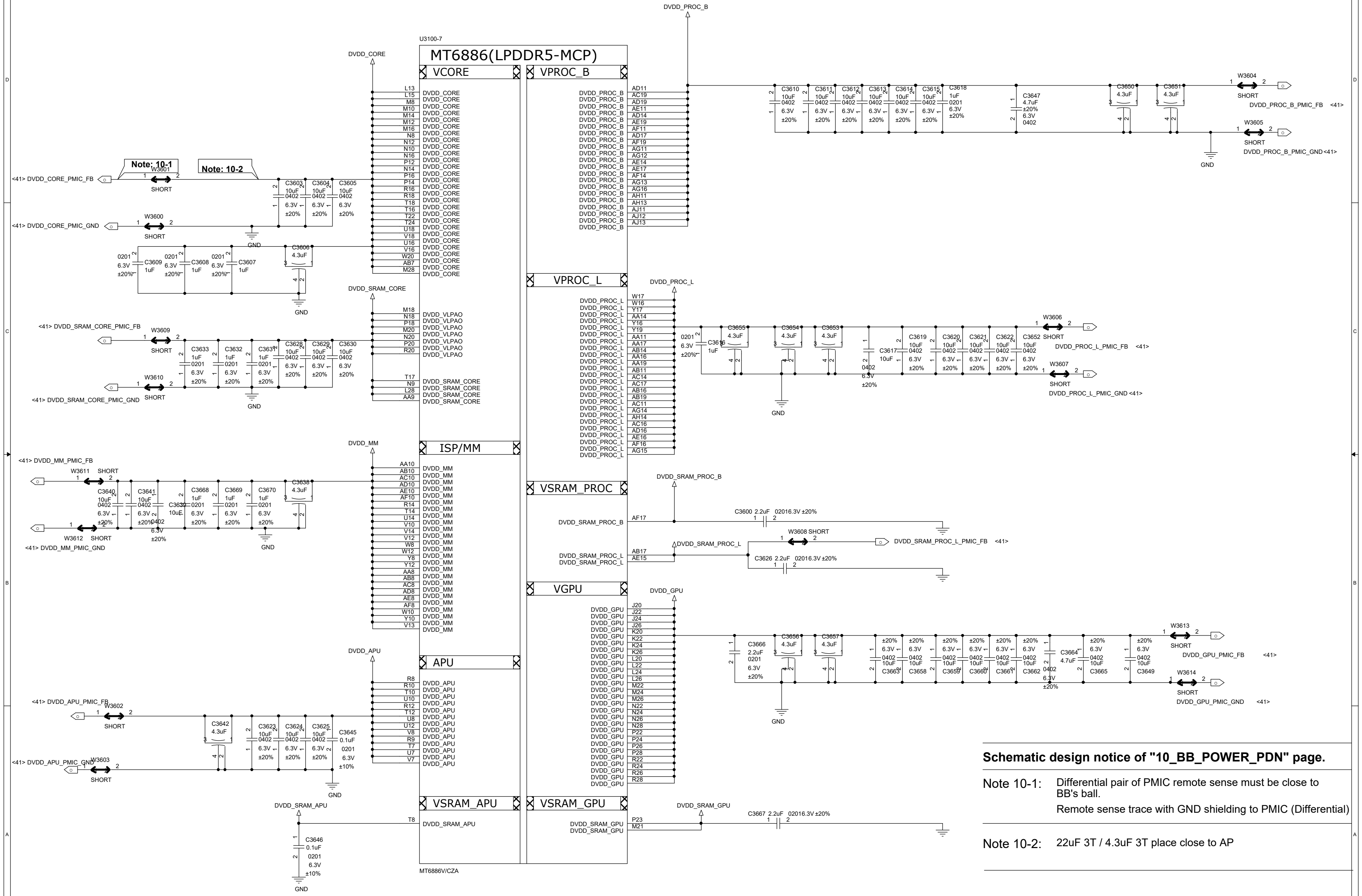
Note 13-3: "AUD_NLE_MOSI0" pins is PMIC strapping for DDR

AUD_NLE_MOSI0	MODE
L (Default)	LP4X
H (by external PU)	LP5

Title	N16U_MAIN
Size A2	Document Number MT6886_INTF
Date: Tuesday, July 18, 2023	Sheet 34 of 73



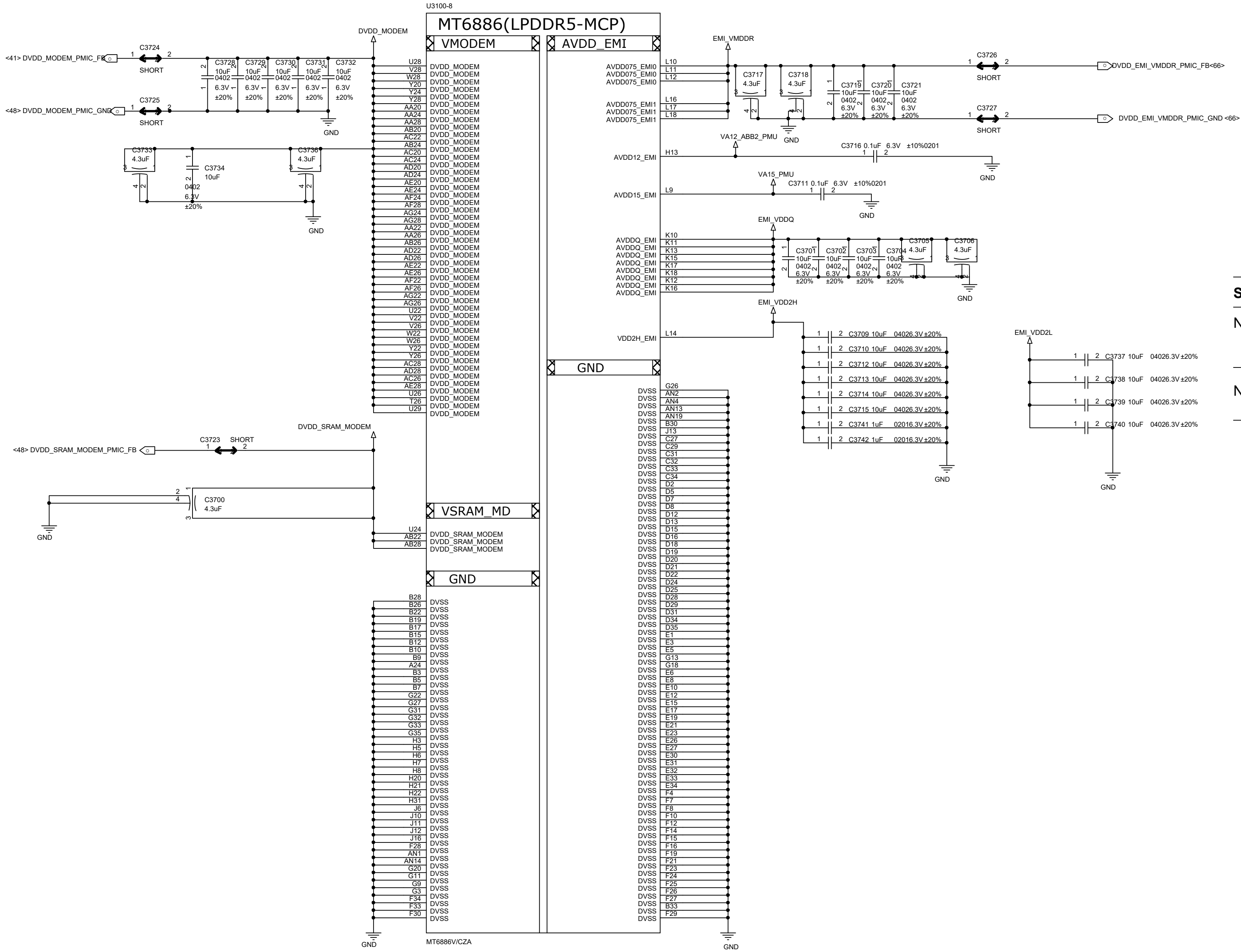
Title	N16U_MAIN
Size C	Document Number MT6886_RF_INTF
Date: Tuesday, July 18, 2023	Sheet 35 73



Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1:	Differential pair of PMIC remote sense must be close to BB's ball. Remote sense trace with GND shielding to PMIC (Differential)
Note 10-2:	22uF 3T / 4.3uF 3T place close to AP

Title: N16U_MAIN
Size: C
Document Number: MT6886_PWR1
Date: Tuesday, July 18, 2023 Sheet 36 of 73



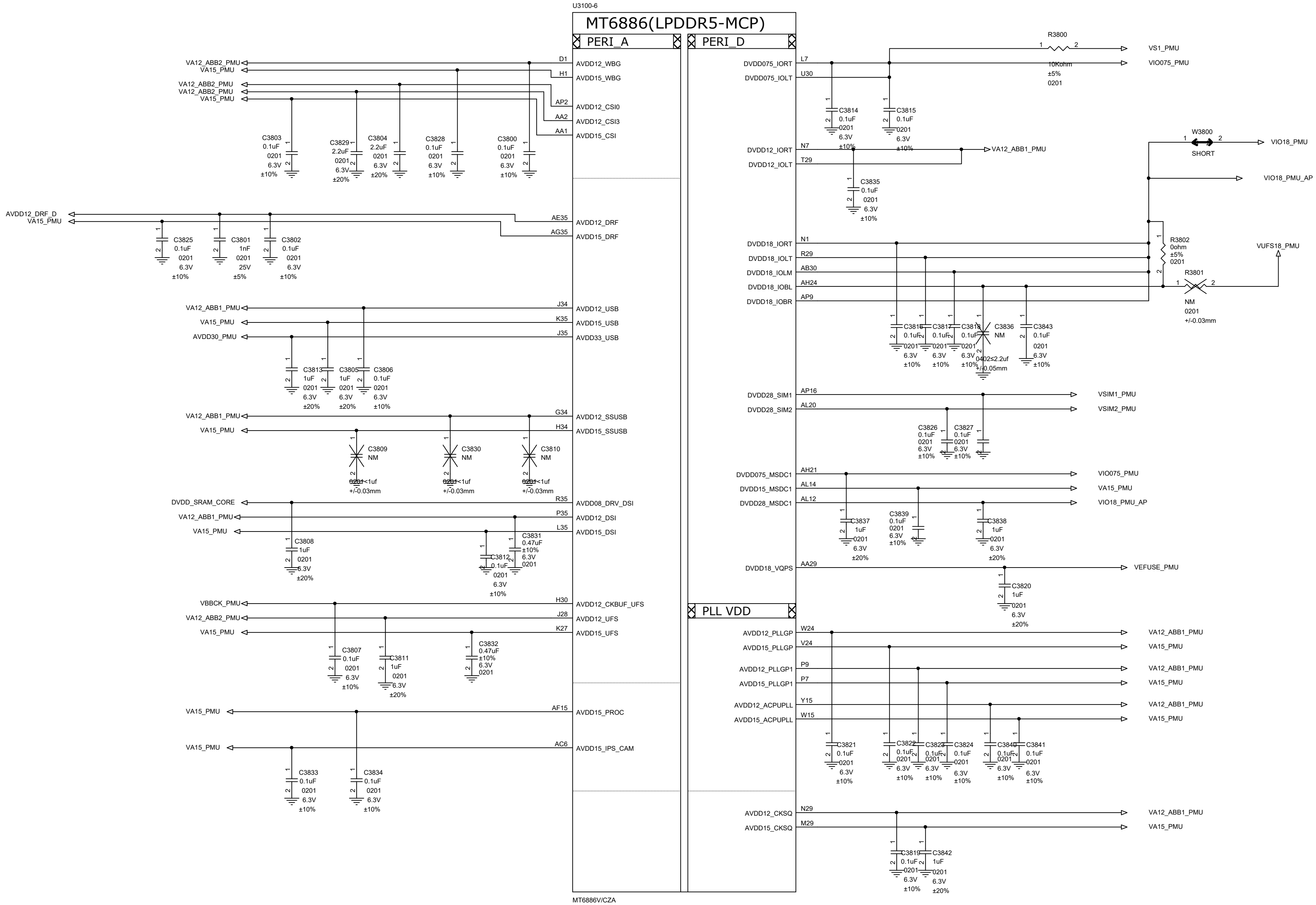
Schematic design notice of "11_BB_POWER_PDN" page.

Note 11-1: Differential pair of PMIC remote sense must be close to BB's ball.

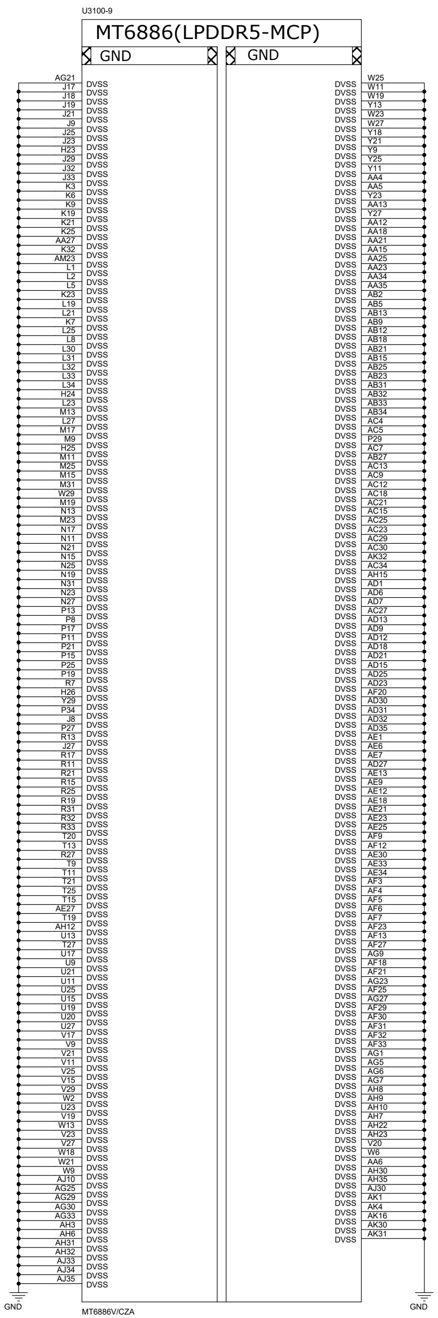
Remote sense trace with GND shielding to PMIC (Differential)

Note 11-2: 22uF 3T / 4.3uF 3T place close to AP

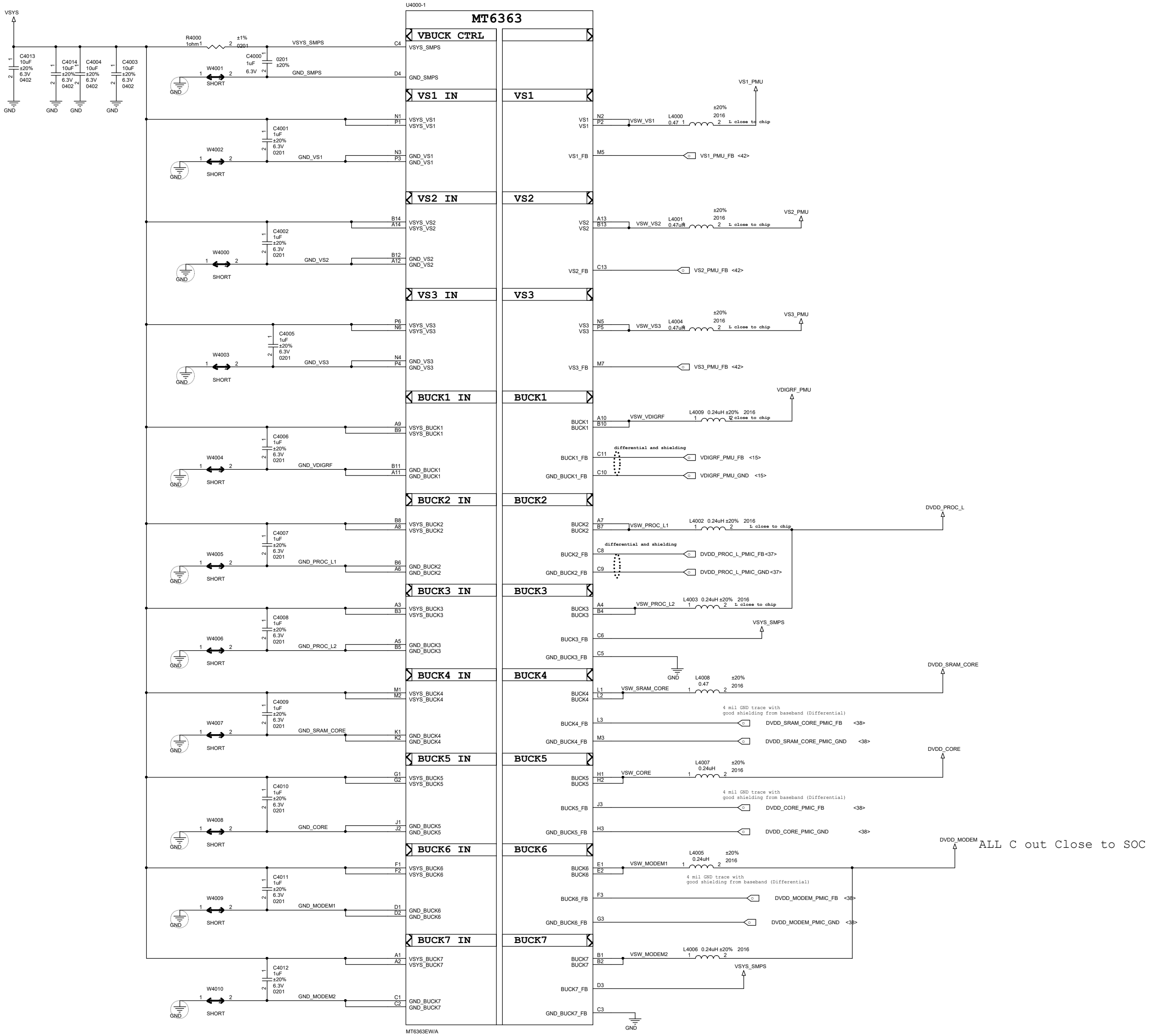
Title	N16U_MAIN
Size A2	Document Number MT6886_PWR2
Date: Tuesday, July 18, 2023	Sheet 37 73



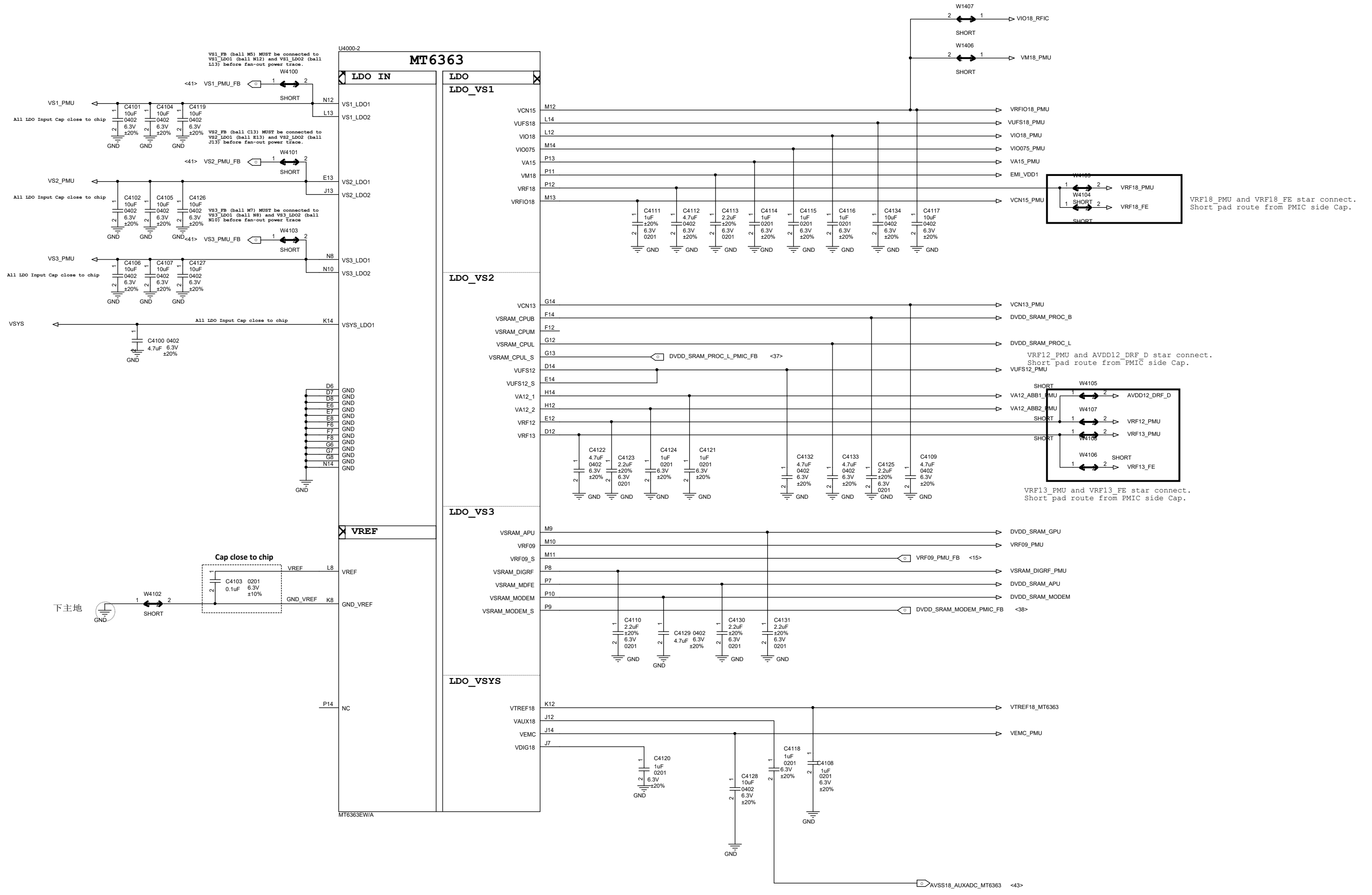
Title	N16U_MAIN
Size A2	Document Number MT6886_PWR3
Date: Tuesday, July 18, 2023	Sheet 38 73



Title	N16U_MAIN		
Size C	Document Number MT6886_GND		
Date:	Tuesday, July 18, 2023	Sheet 39	of 73



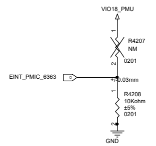
Title	N16U_MAIN
Size A1	Document Number MT6363_BUCK
Date: Tuesday, July 18, 2023	Sheet

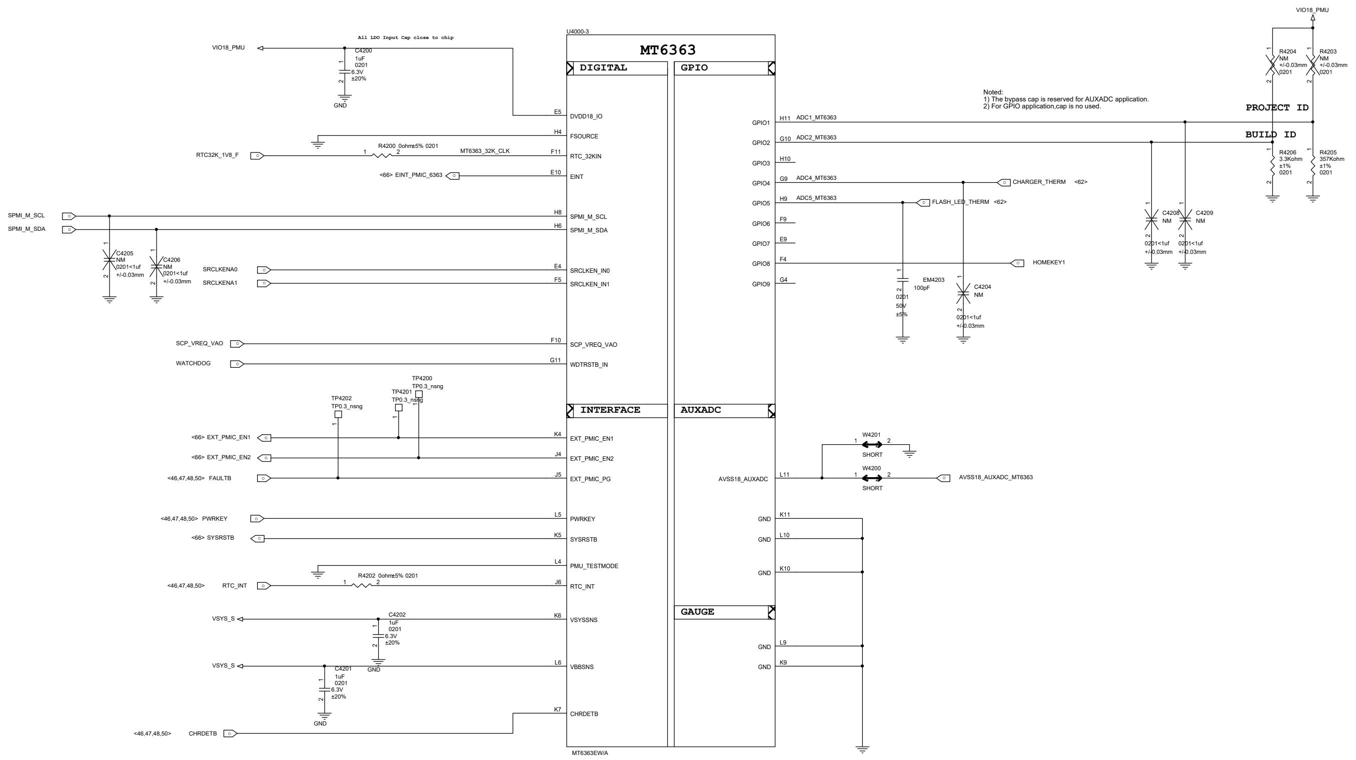


Title	N16U_MAIN
Size A1	Document Number MT6363_LDO
Date: Tuesday, July 18, 2023	Sheet 41 of 73

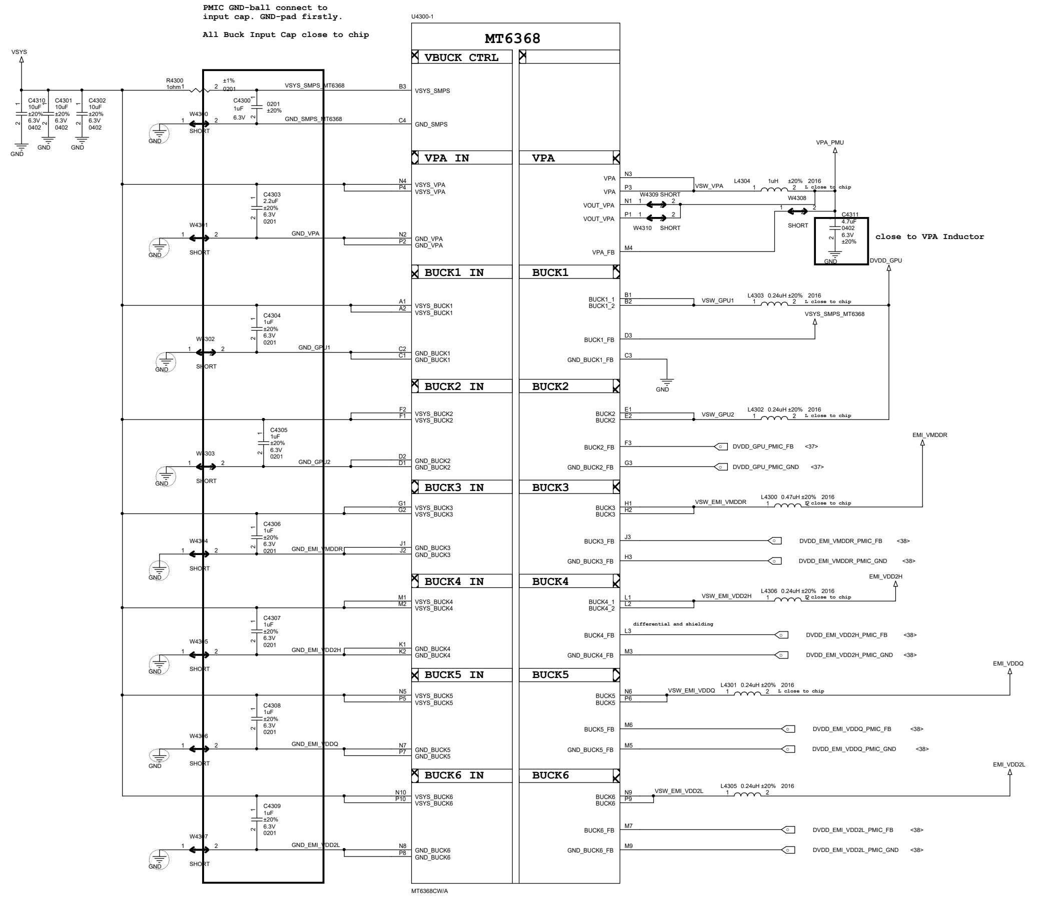
MT6363	Pin : EINT	VEMC
UFS3.1	0	2.55V
EMMC/UFS2.2	1	3V

General MT6363
VEMC selection HW TRAPPING

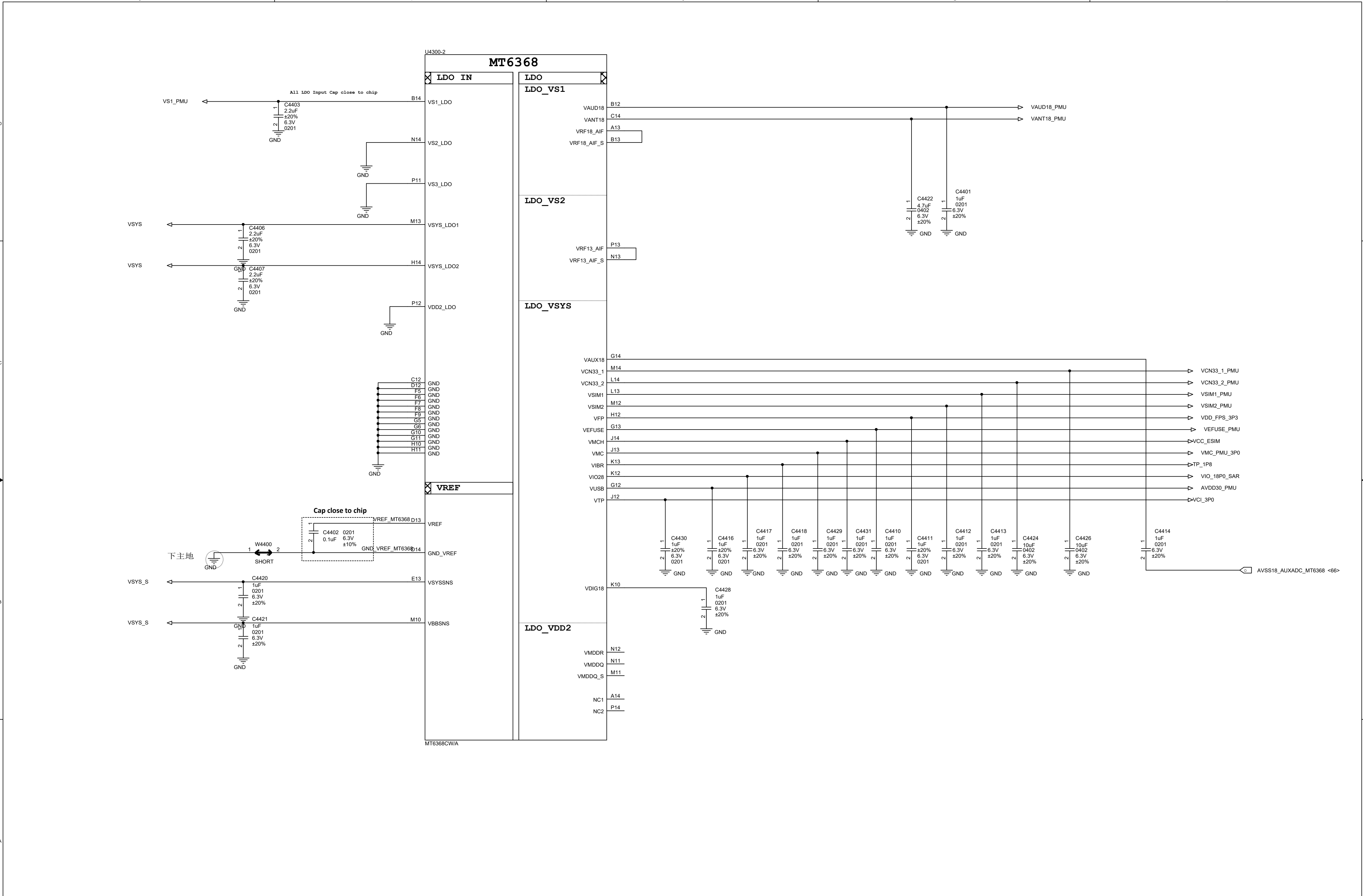




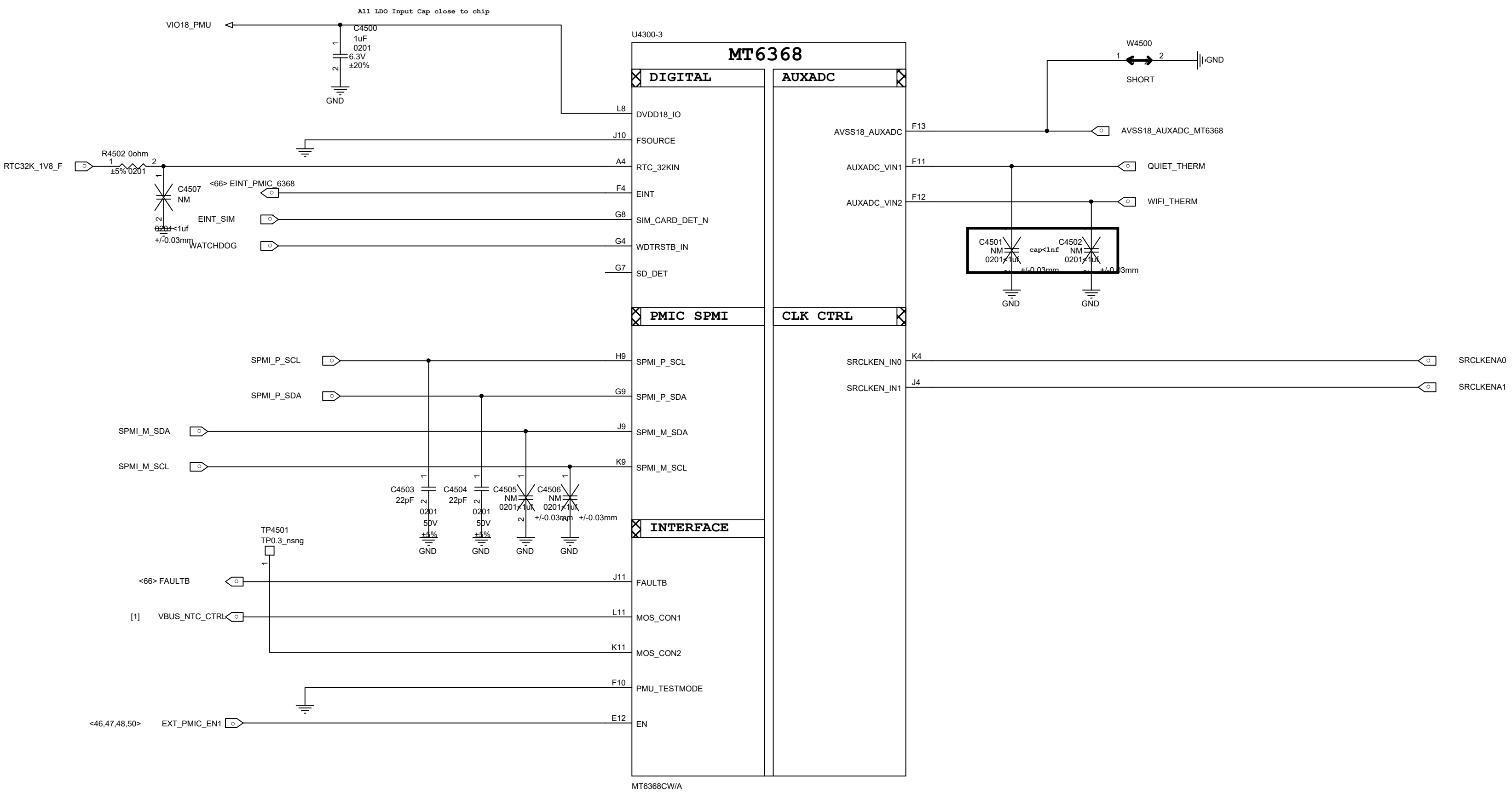
Title	N16U_MAIN		
Size A1	Document Number MT6363_General		
Date:	Tuesday, July 18, 2023	Sheet 42	73



Title	N16U_MAIN
Size A1	Document Number MT6368_BUCK
Date: Tuesday, July 18, 2023	Sheet 43 of 73



Title	N16U_MAIN
Size A2	Document Number MT6368_LDO
Date: Tuesday, July 18, 2023	Sheet 44 73

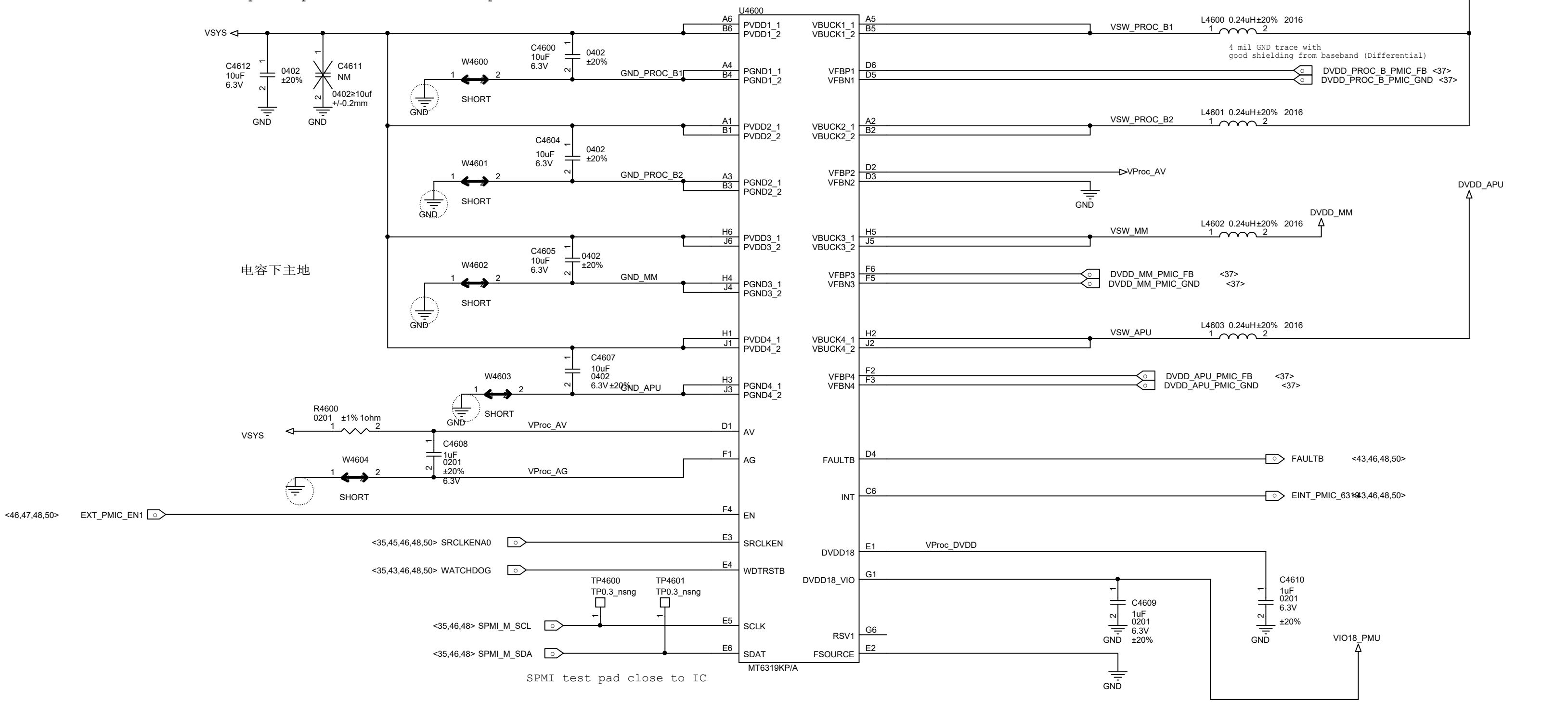


Title	N16U_MAIN
Size A2	Document Number MT6368_Gerneral
Date: Tuesday, July 18, 2023	Sheet 45 73

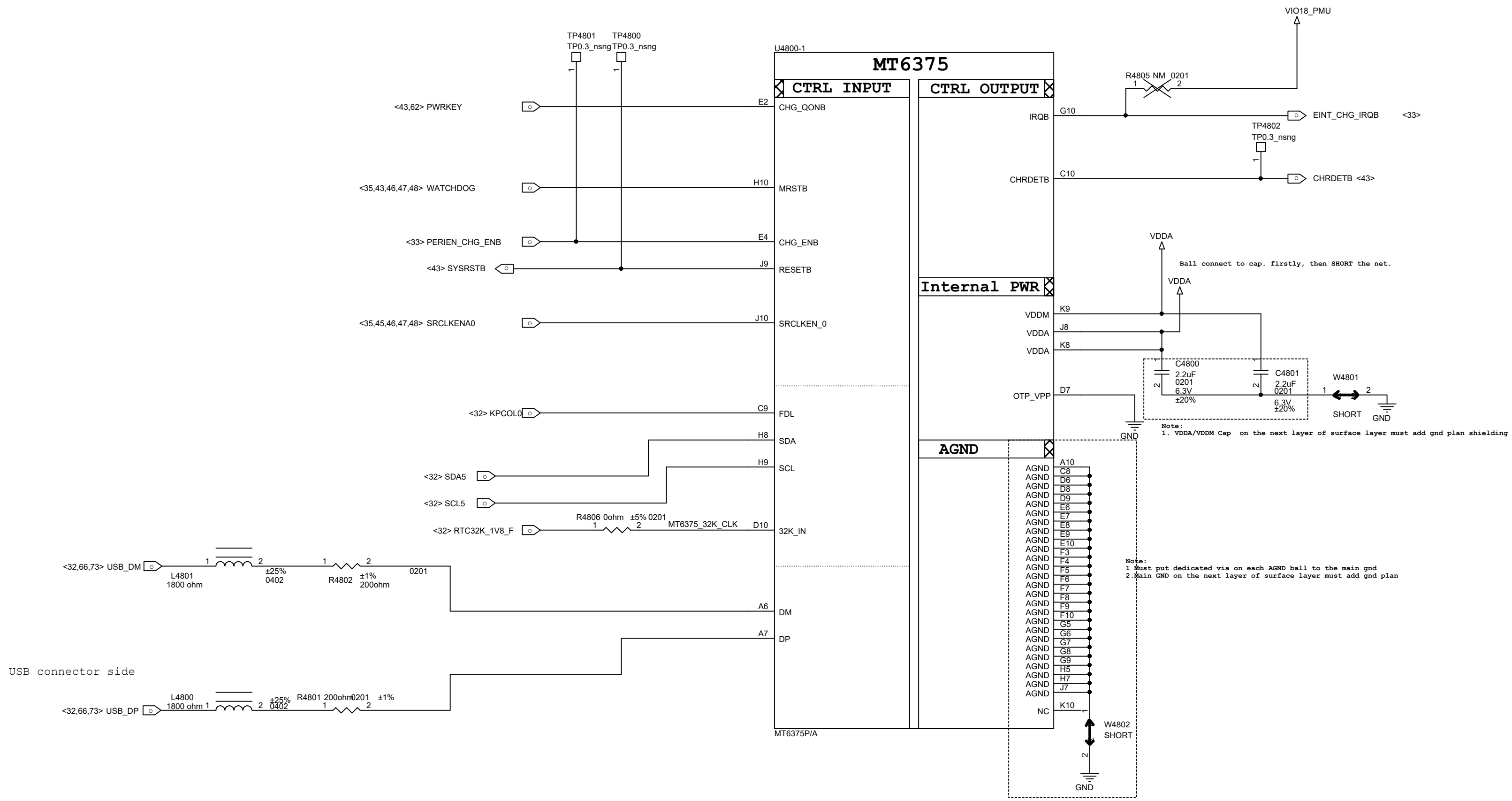
MT6319 4-Phase Buck (CPU)

ALL C out Close to SOC

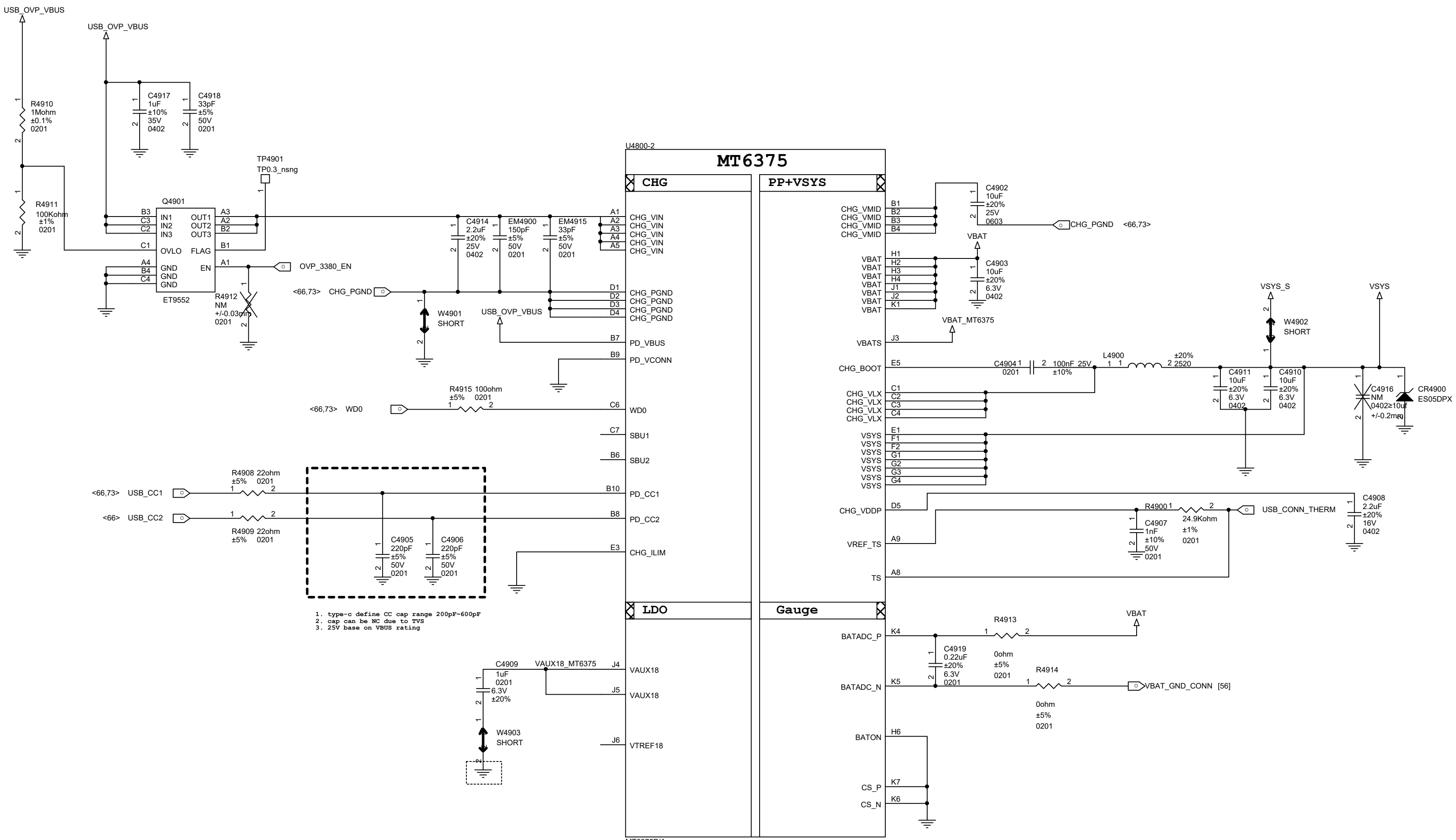
Input cap GND connect to chip first than to SYS GND



Title	N16U_MAIN
Size A2	Document Number MT6319_VCPUB
Date: Tuesday, July 18, 2023	Sheet 46 of 73

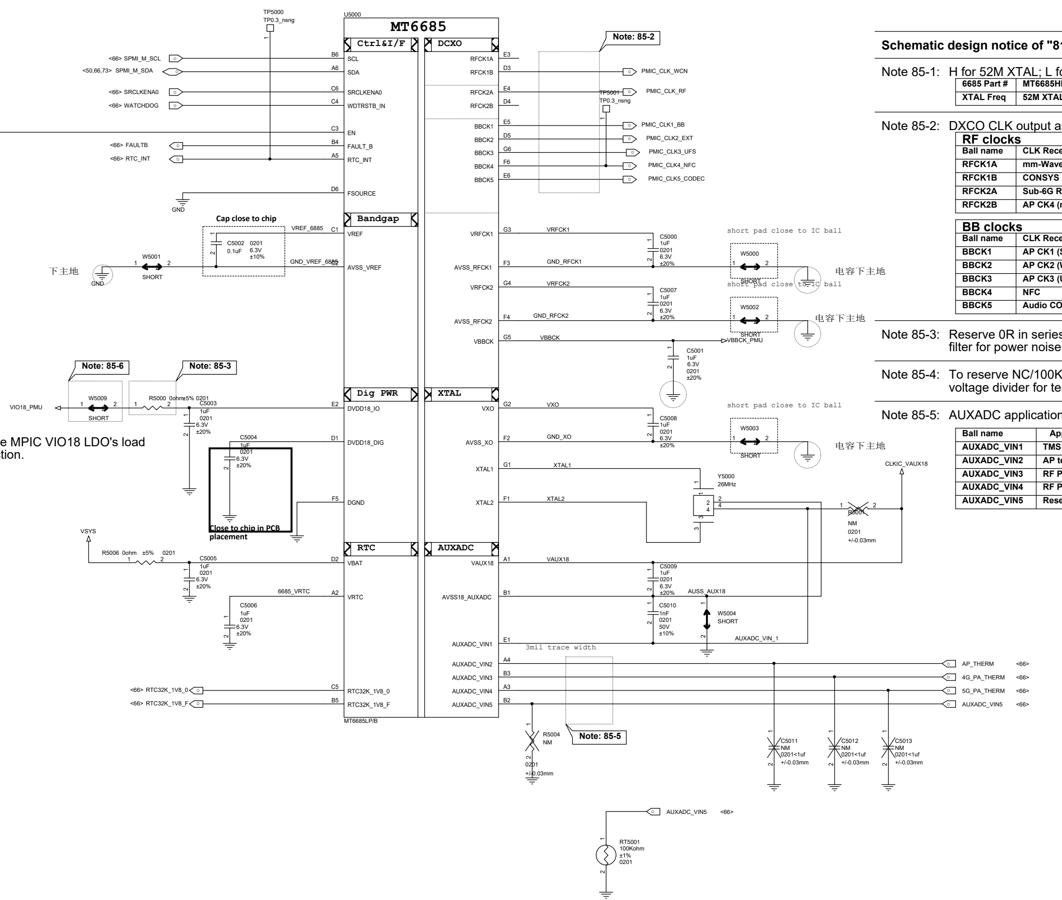


Title	N16U_MAIN
Size A2	Document Number MT6375_General
Date: Tuesday, July 18, 2023	Sheet 48 of 73



Note: VAUX18/VTREF18 Cap on the next layer of surface layer must add gnd plan shielding

Title	N16U_MAIN
Size A2	Document Number MT6375_CHARGER
Date: Tuesday, July 18, 2023	Sheet 49 of 73



Note 85-6:

Short pad must be placed on the MPIC VIO18 LDO's load cap. side for power star-connection.

Schematic design notice of "81_MT6685 CLK AUXADC" page.

Note 85-1: H for 52M XTAL; L for H for 26M XTAL

6685 Part #	MT6685HP/A	MT6685LP/A
XTAL Freq	52M XTAL	26M XTAL

Note 85-2: DXCO CLK output assignment to CLK receiver

RF clocks

Ball name	CLK Receiver	Freq.(MHz)
RFCK1A	mm-Wave RF	XTAL (26/52)
RFCK1B	CONSYS RF	XTAL (26/52)
RFCK2A	Sub-6G RF	26M
RFCK2B	AP CK4 (mm-Wave ABB)	26M

BB clocks

Ball name	CLK Receiver	Freq.(MHz)
BBCK1	AP CK1 (System-boot)	26M
BBCK2	AP CK2 (WBG)	26M
BBCK3	AP CK3 (UFS)	26M
BBCK4	NFC	26M
BBCK5	Audio CODEC	26M

Note 85-3: Reserve 0R in series connection to form a R/C low pass filter for power noise reduction

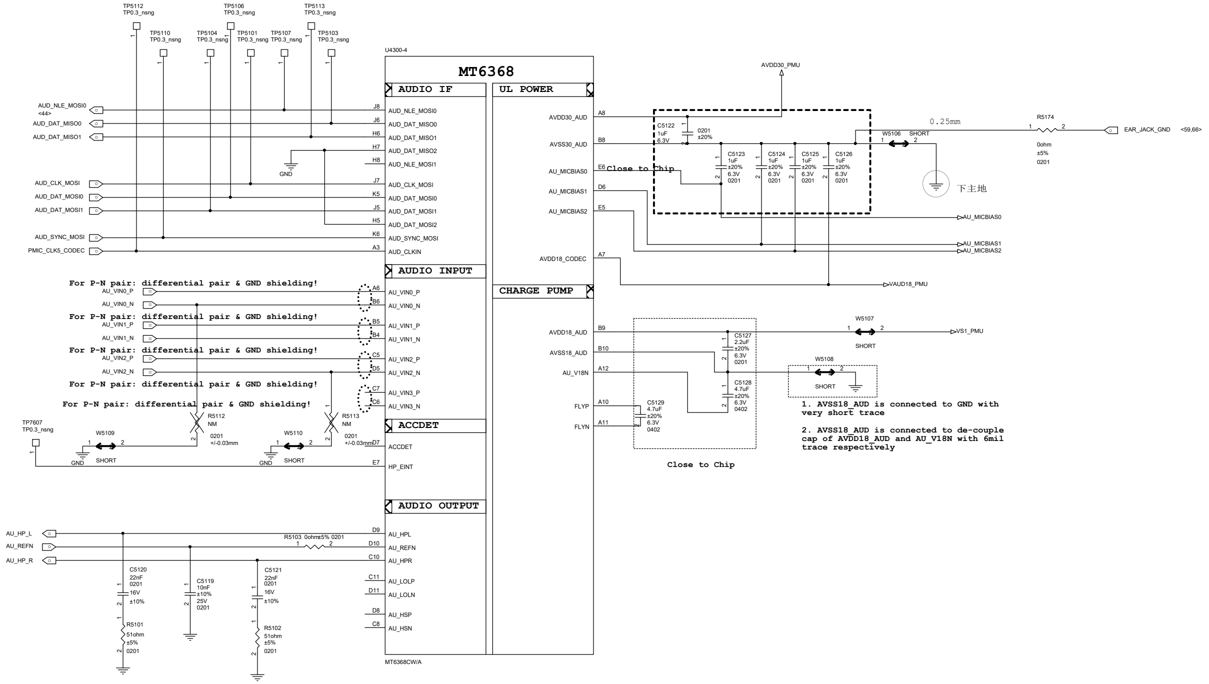
Note 85-4: To reserve NC/100K ohm(1%) on AUXADC_VIN1 as voltage divider for temperature sensor in TMS

Note 85-5: AUXADC application

Ball name	Application
AUXADC_VIN1	TMS
AUXADC_VIN2	AP temperature measurement in board level
AUXADC_VIN3	RF PA1 temperature measurement
AUXADC_VIN4	RF PA2 temperature measurement
AUXADC_VIN5	Reserved. (Tied to GND for not used AUXADC)

Placed on the hear source which is nearest to Y5000. Eg: PMIC, Charger, CPU, WIFI_PA.

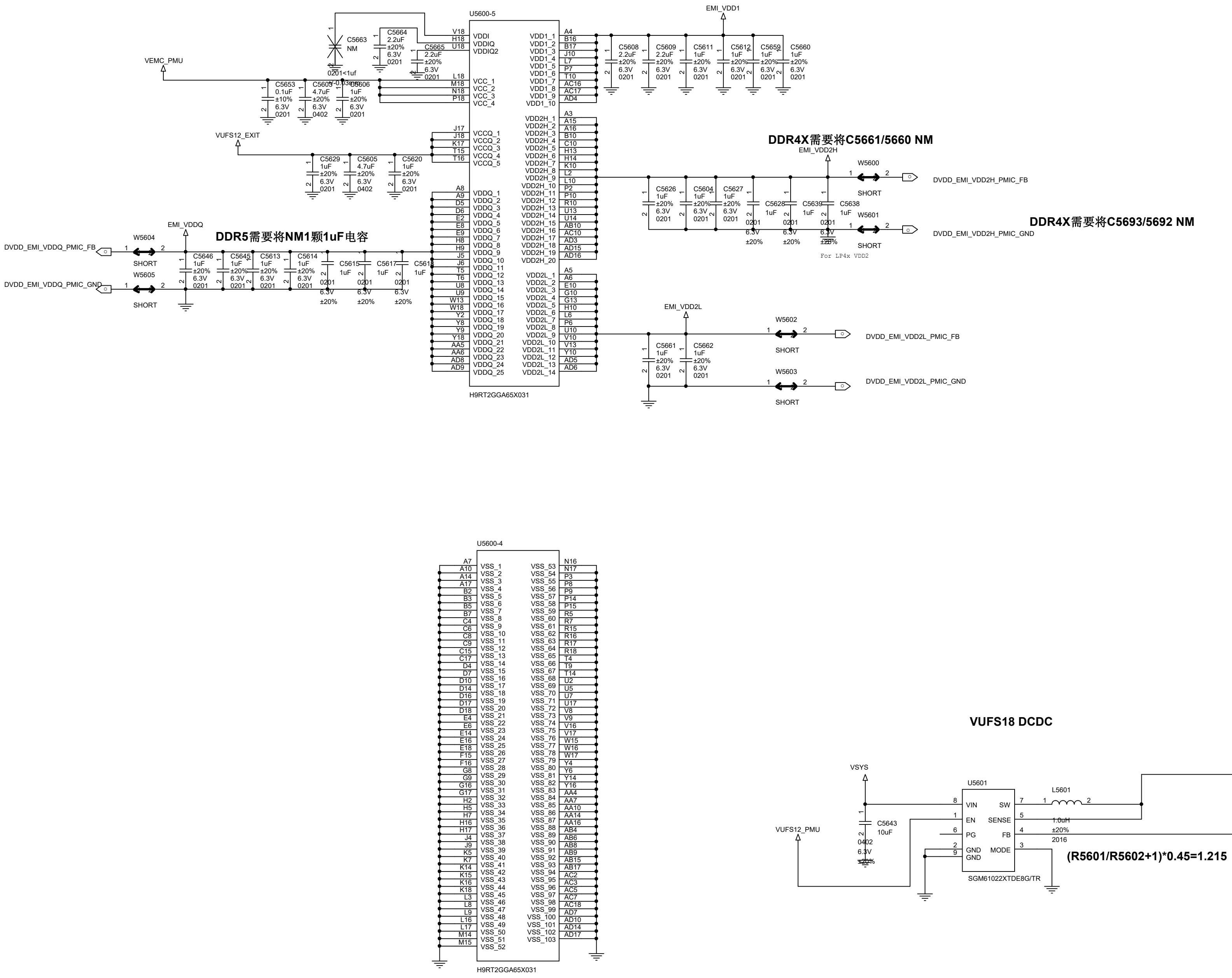
Title	N16U_MAIN
Size A1	Document Number MT6685_CLK
Date: Tuesday, July 18, 2023	Sheet 50 of 73



下主地

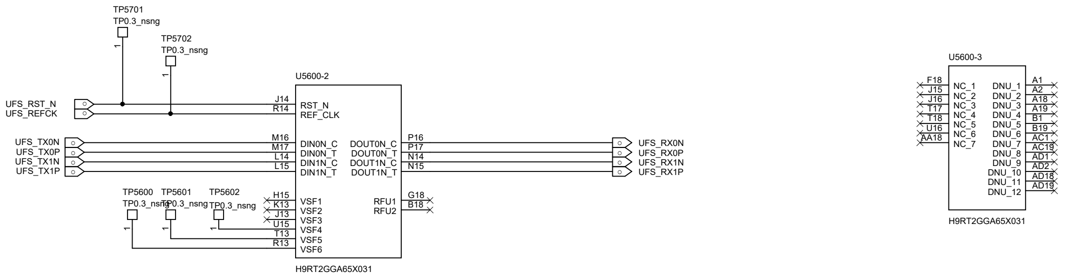
-AU_HPL and AU_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively

-The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"

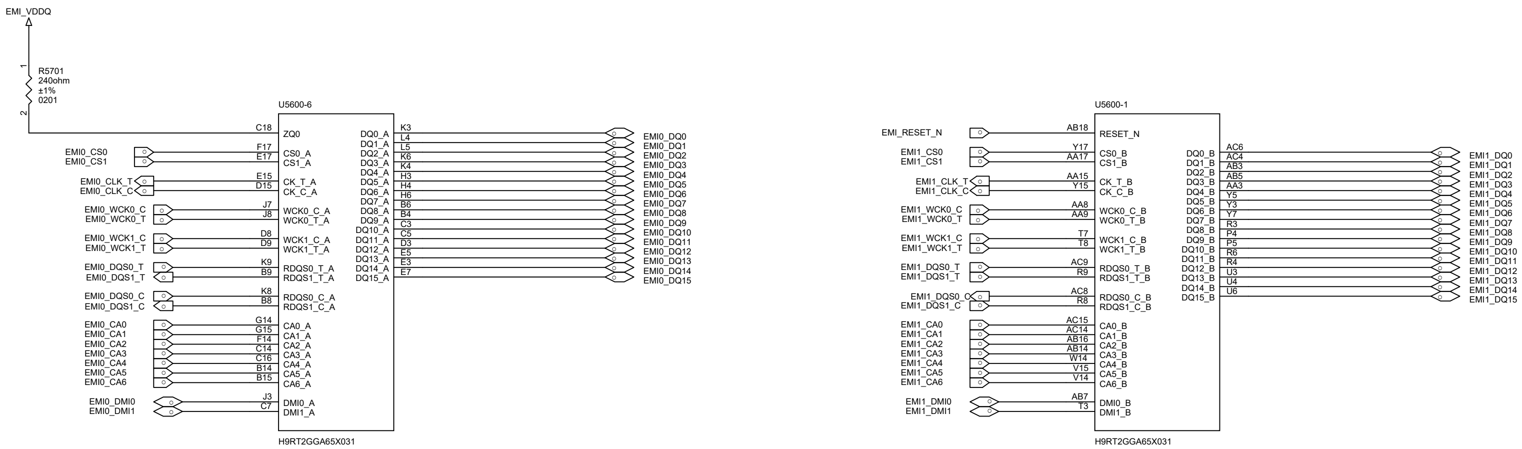


Title	N16U_MAIN
Size A2	Document Number MEMORY_1
Date: Tuesday, July 18, 2023	Sheet

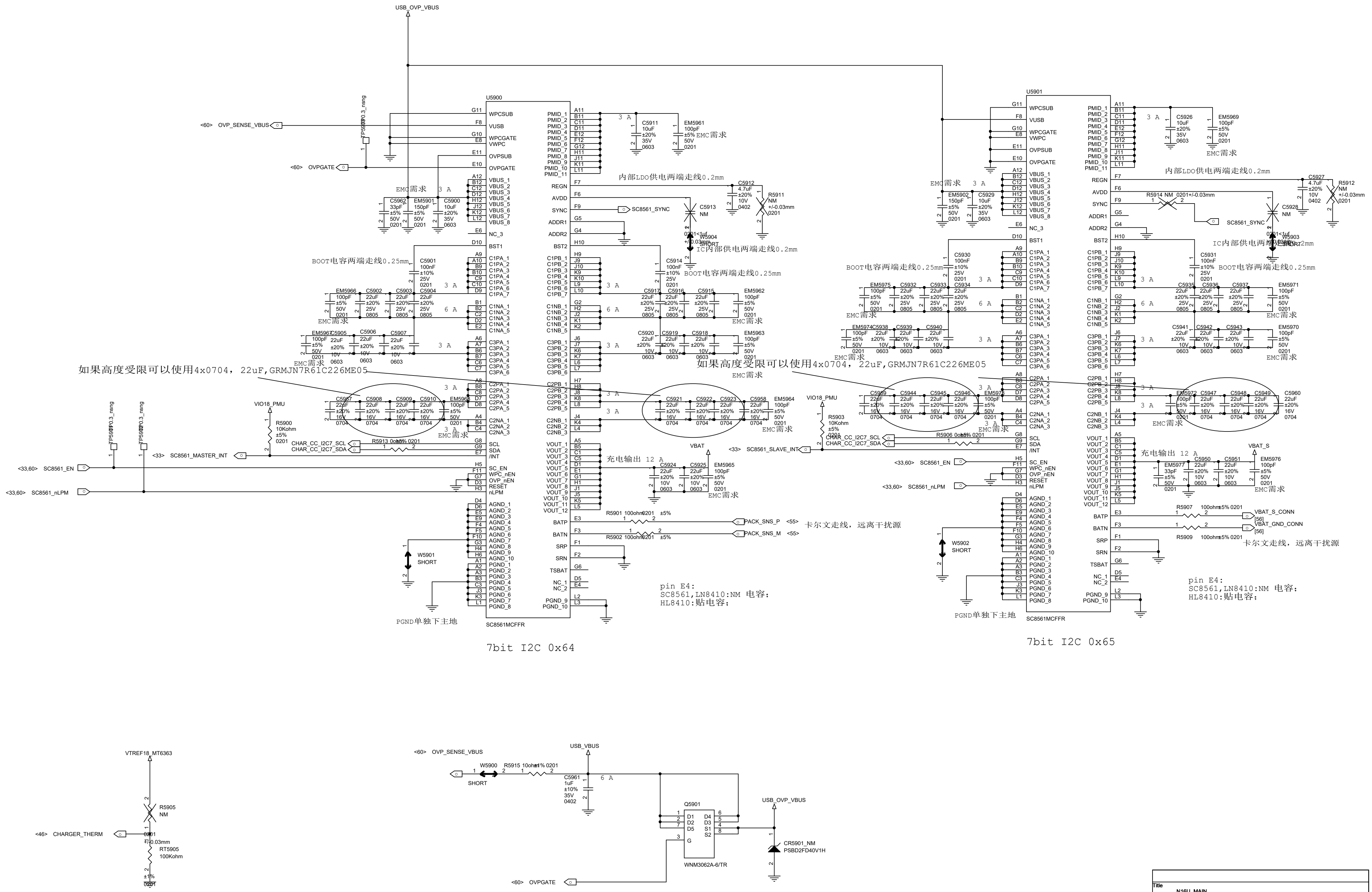
UFS 3.1



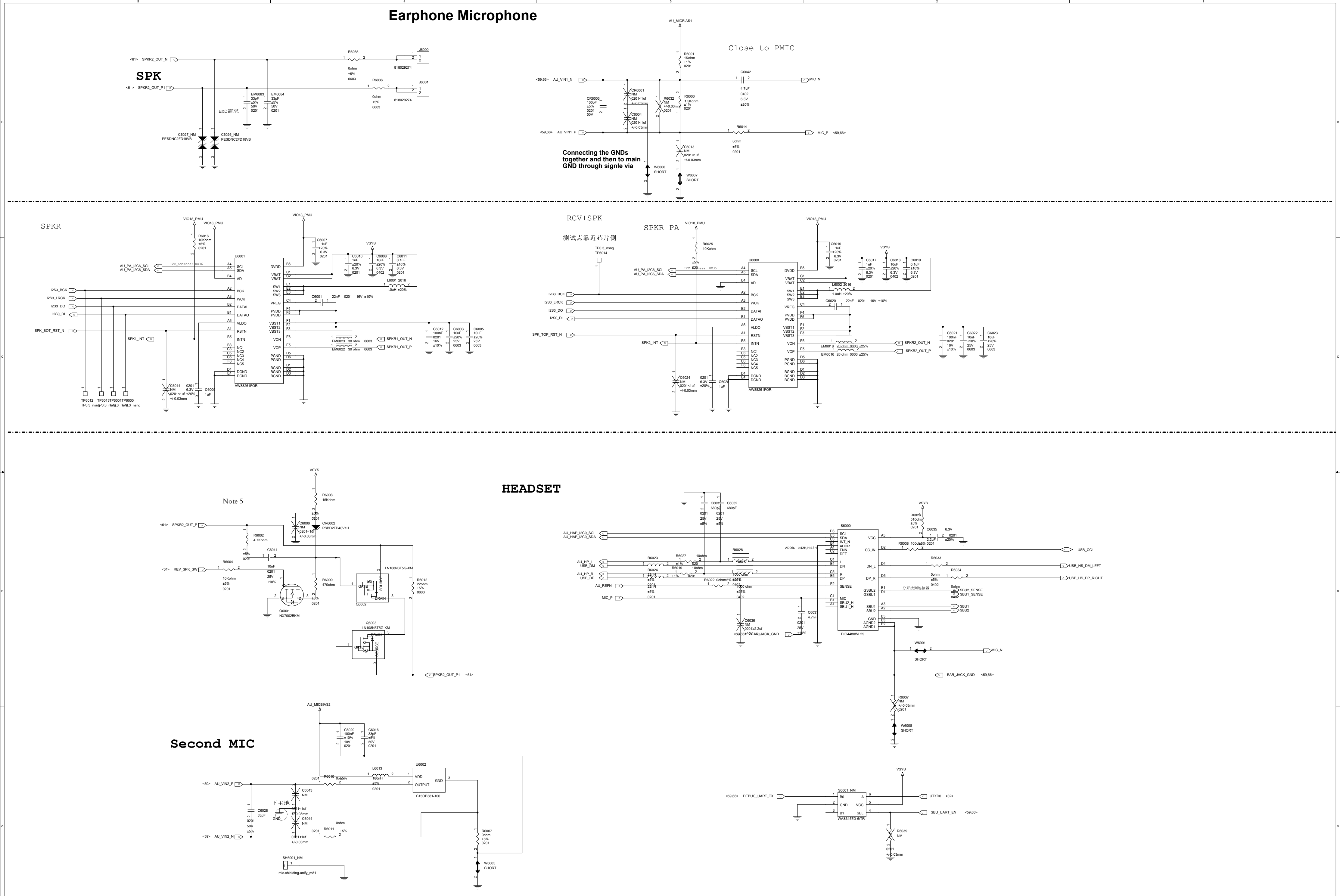
LPDDR5



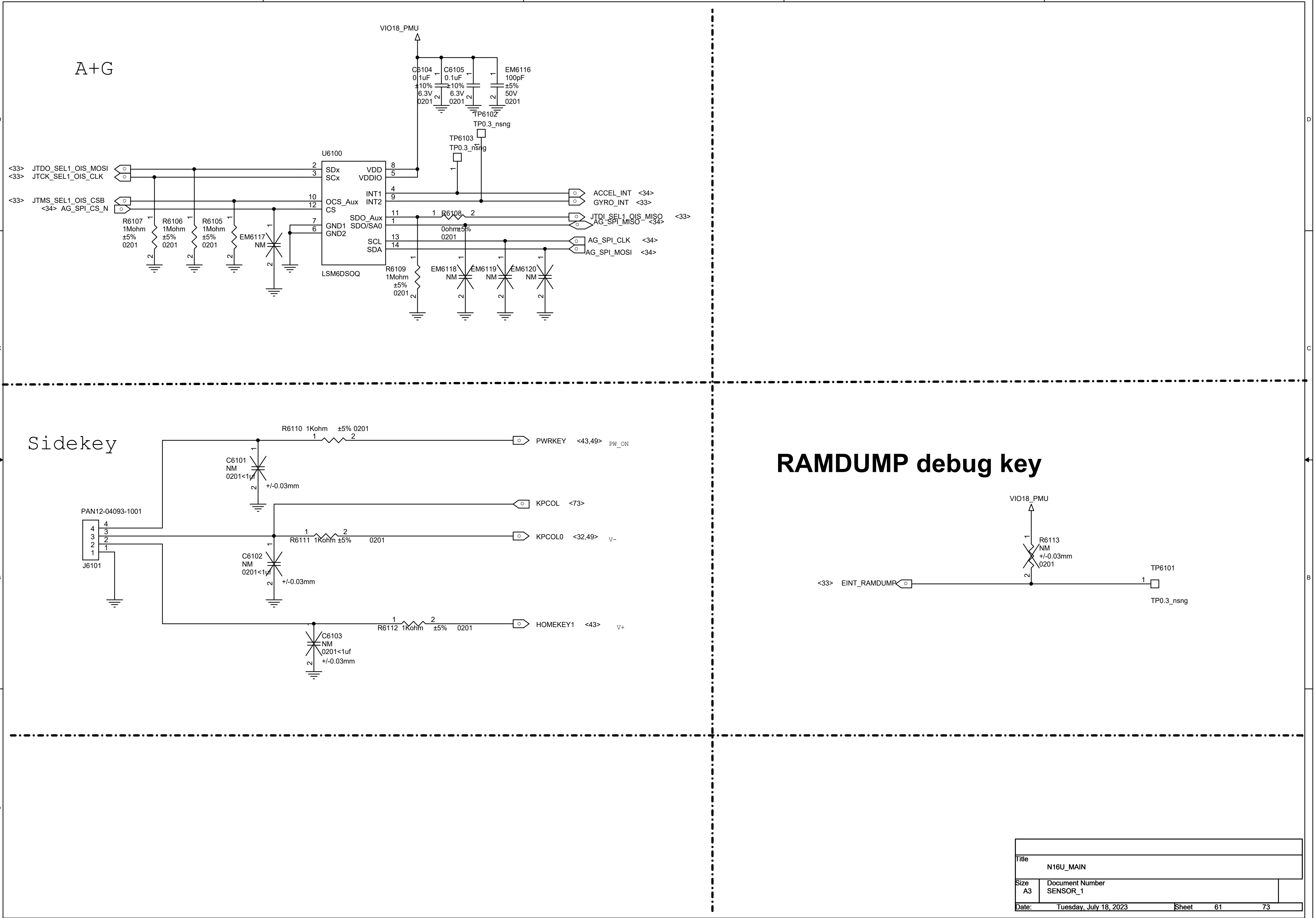
Title	N16U_MAIN
Size A2	Document Number MEMORY_2
Date: Tuesday, July 18, 2023	Sheet 57 of 73



Title	N16U_MAIN
Size A2	Document Number SC8561
Date: Tuesday, July 18, 2023	Sheet 59 of 73



Title	N16U_MAIN
Size A1	Document Number AUDIO
Date: Tuesday, July 18, 2023	Sheet 60 of 73


A+G
VIO18_PMU
C6104 0.1uF ±10% 6.3V 0201
C6105 0.1uF ±10% 6.3V 0201
EM6116 100pF ±5% 50V 0201
TP6102 TP0.3_nsng
TP6103 TP0.3_nsng
U6100
LSM6DSOQ
SDx SCx OCS_Aux CS GND1 GND2
VDD VDDIO INT1 INT2 SDO_Aux SDO/SA0 SCL SDA
<33> JTDO_SEL1_OIS_MOSI
<33> JTCK_SEL1_OIS_CLK
<33> JTMS_SEL1_OIS_CSB
<34> AG_SPI_CS_N
R6107 1Mohm ±5% 0201
R6106 1Mohm ±5% 0201
R6105 1Mohm ±5% 0201
EM6117 NM
ACCEL_INT <34>
GYRO_INT <33>
R6108 0ohm ±5% 0201
JTDI_SEL1_OIS_MISO <33>
AG_SPI_MISO <34>
AG_SPI_CLK <34>
AG_SPI_MOSI <34>
R6109 1Mohm ±5% 0201
EM6118 NM
EM6119 NM
EM6120 NM
Sidekey
R6110 1Kohm ±5% 0201
PWRKEY <43,49> PW_ON
C6101 NM 0201<1uf +/-0.03mm
KPCOL <73>
PAN12-04093-1001
J6101
R6111 1Kohm ±5% 0201
KPCOL0 <32,49> V-
C6102 NM 0201<1uf +/-0.03mm
R6112 1Kohm ±5% 0201
HOMEKEY1 <43> V+
C6103 NM 0201<1uf +/-0.03mm
RAMDUMP debug key
VIO18_PMU
R6113 NM +/-0.03mm 0201
<33> EINT_RAMDUMP
TP6101 TP0.3_nsng
Title N16U_MAIN
Size A3
Document Number SENSOR_1
Date: Tuesday, July 18, 2023
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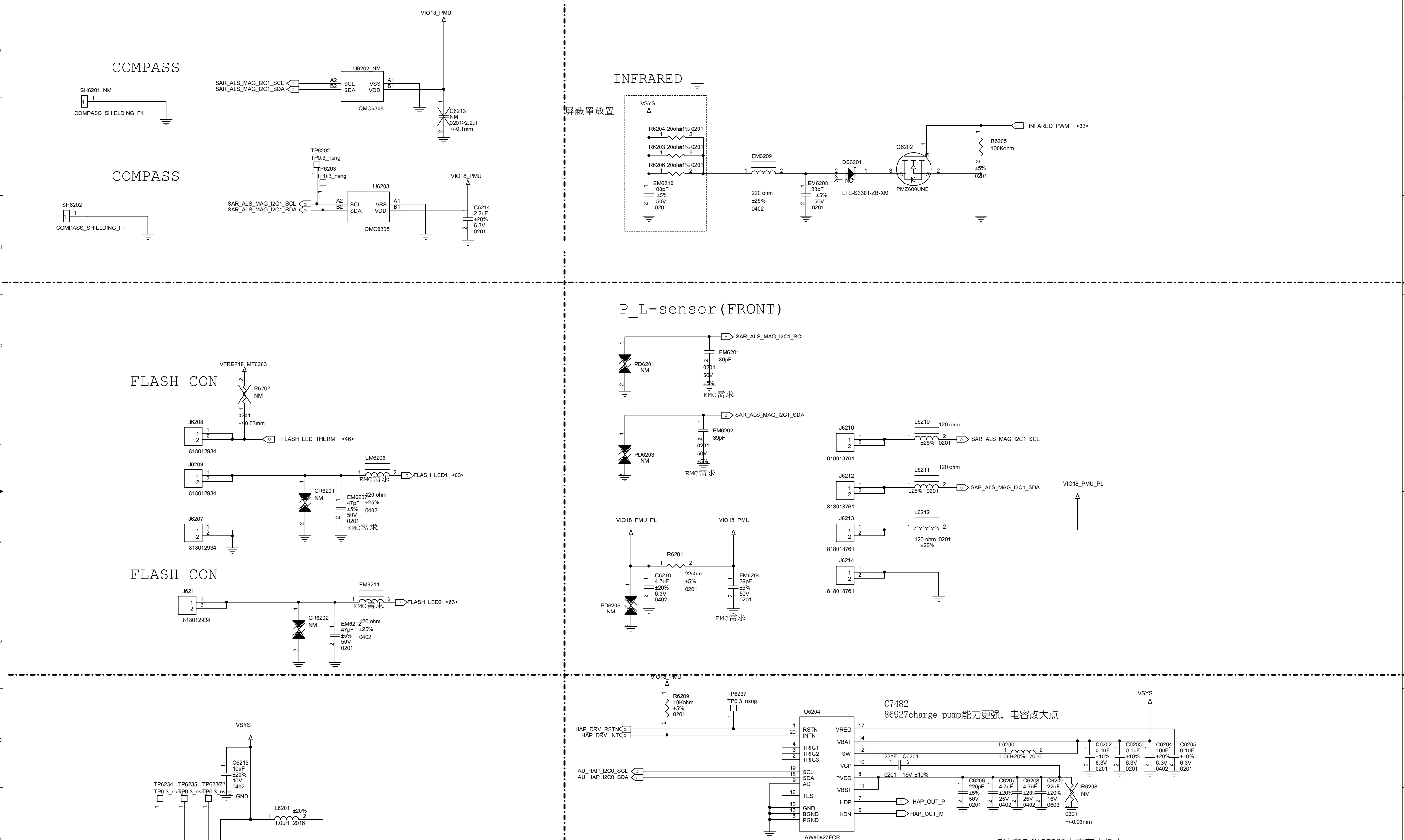
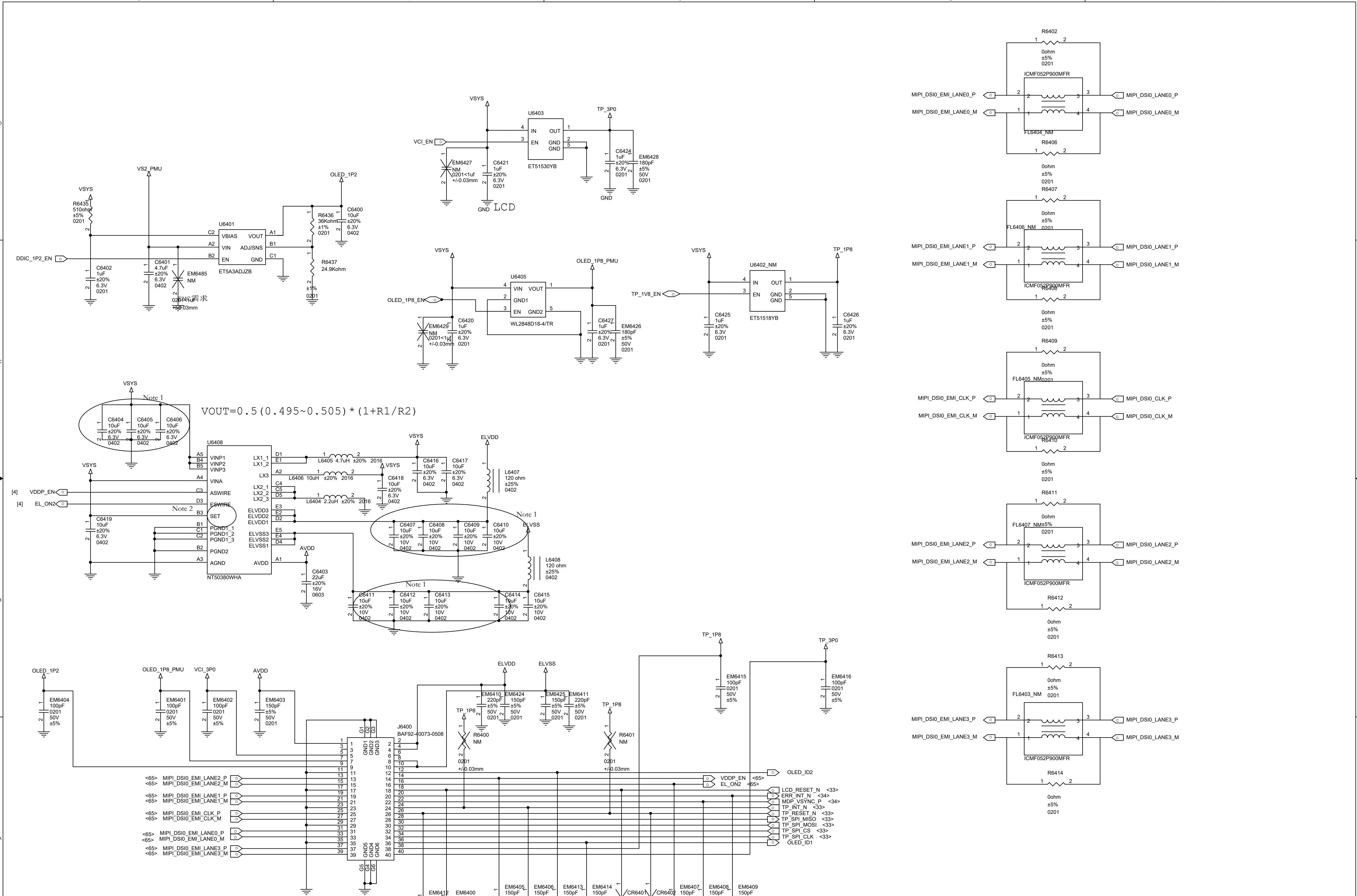


R1 100Kohm是泄放电阻，可以先NM
使用泄放电阻的原因是：
1.扣电池时，GND碰到电池vbat，PVDD电压变成-Vbat电压，且无法泄放，
  再正常连接后，从芯片vbat端到PVDD有瞬间大电流，把mos管烧坏
2.可能会受到打静电影响，L10A没贴，打静电mos管损坏，L10贴了，没损坏过

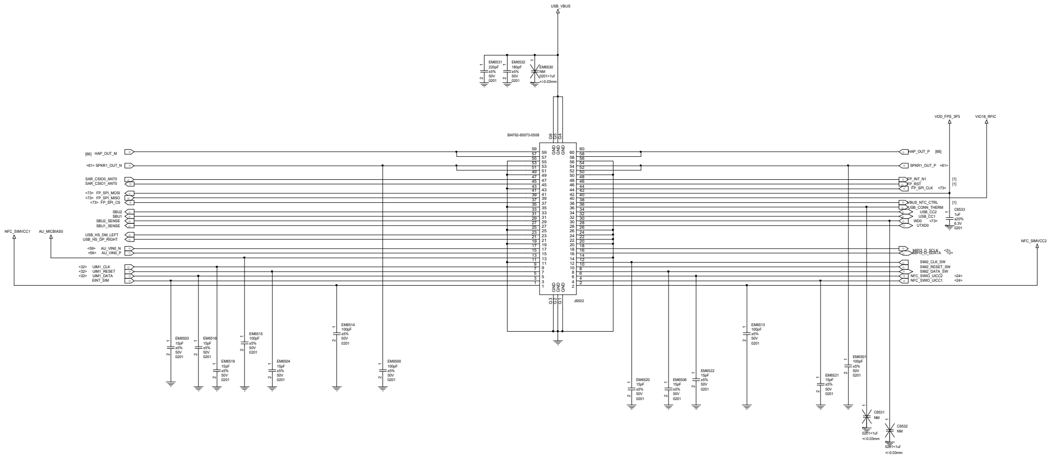
【注意】HAPTICS未来在小板上
L0201 限高1.0以上，使用240100000N79
限高0.8，则使用24V01R0PI1E6

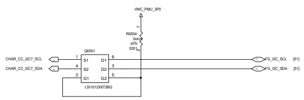
HAP_DRV_INT、HAP_DRV_RSTN还未接

Title	N16U_MAIN
Size A2	Document Number SENSOR_2
Date: Tuesday, July 18, 2023	Sheet

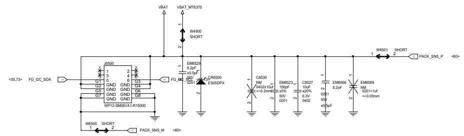


Title	N16U_MAIN
Size A2	Document Number AMOLED
Date: Tuesday, July 18, 2023	Sheet 64 of 73

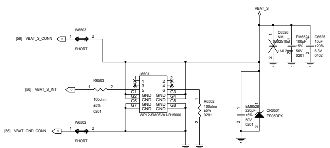




PRIMARY BAT CON

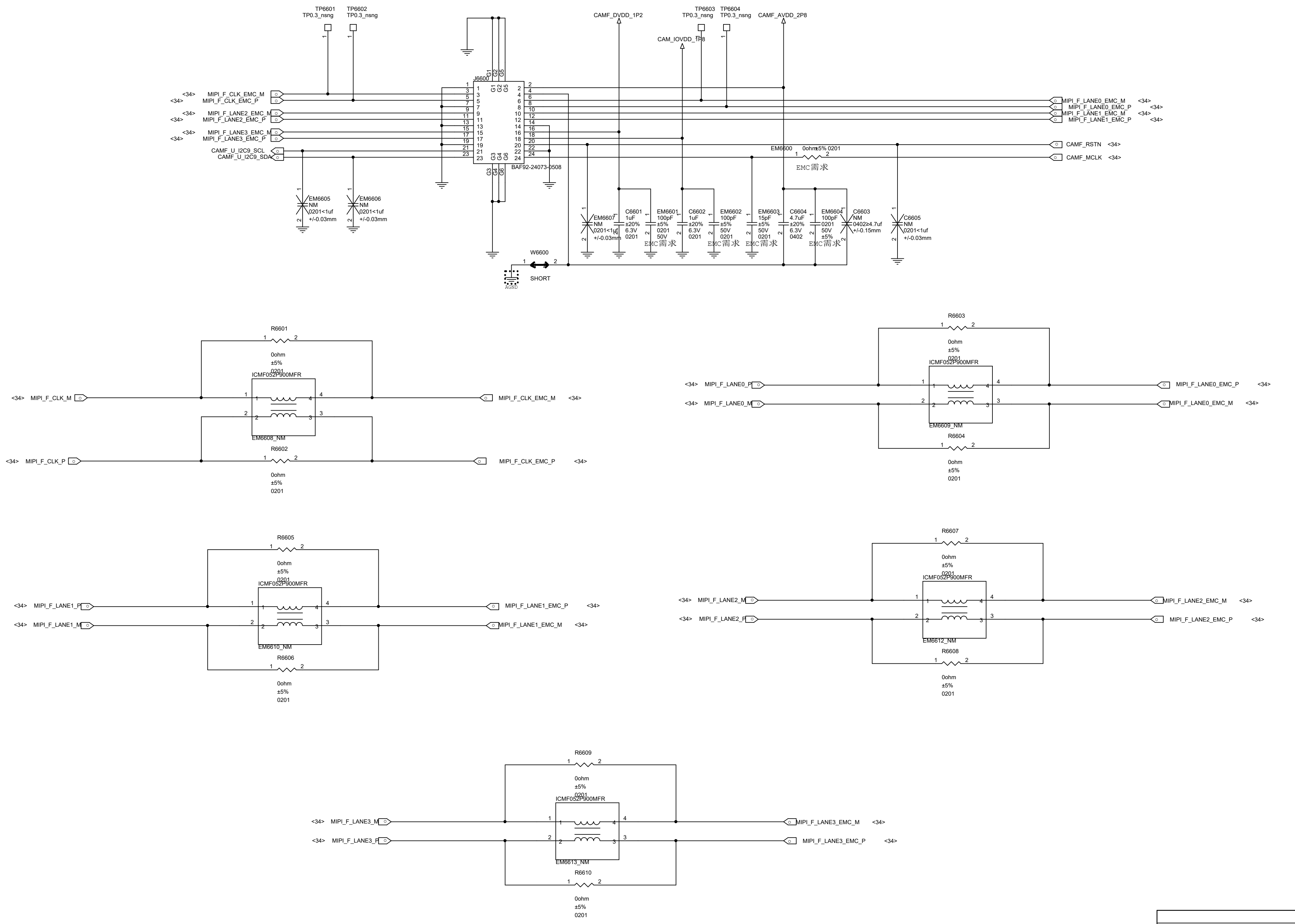


SECONDARY BAT CON



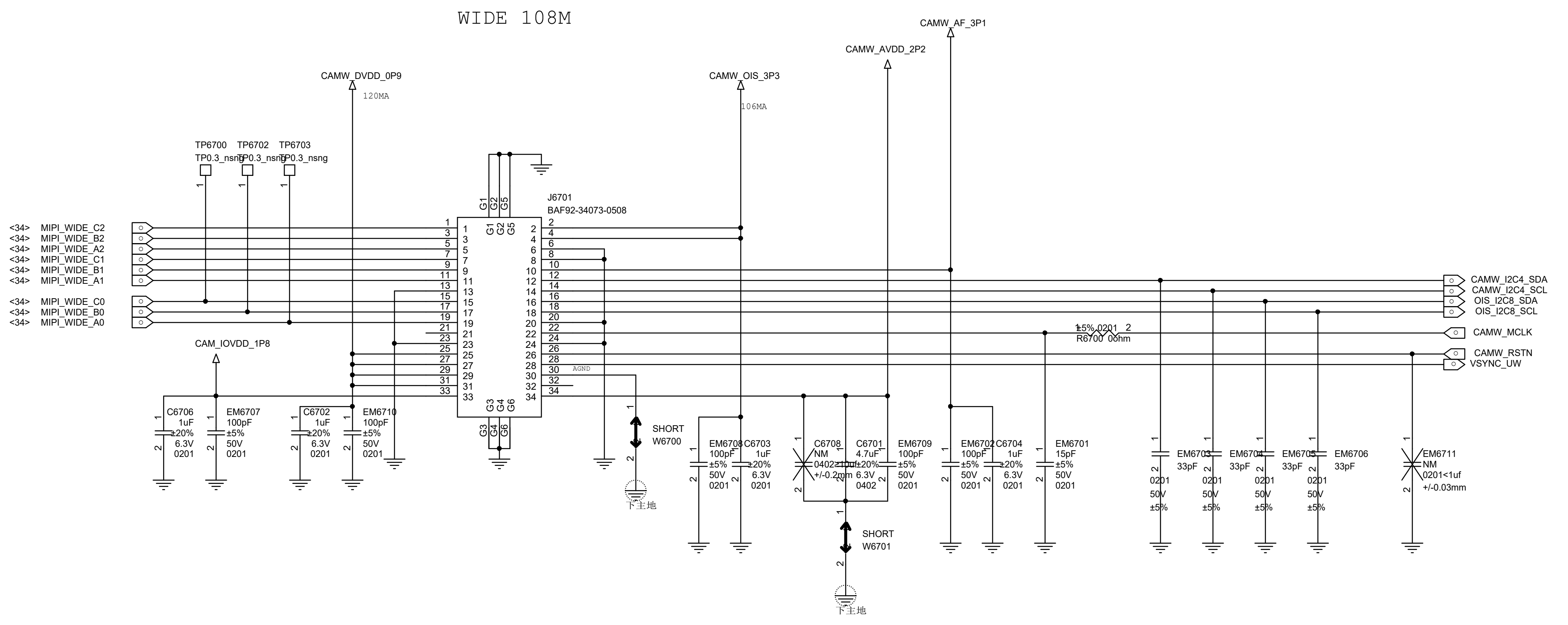
Title	N16U_MAIN
Size A0	Document Number BTB
Date: Tuesday, July 18, 2023	Sheet

front 16M OV16A1Q-GA5A



Title	N16U_MAIN
Size A2	Document Number FRONT_CAMERA
Date: Tuesday, July 18, 2023	Sheet

WIDE 108M(S5KHP3)



Title	N16U_MAIN		
Size A3	Document Number WIDE_CAMERA		
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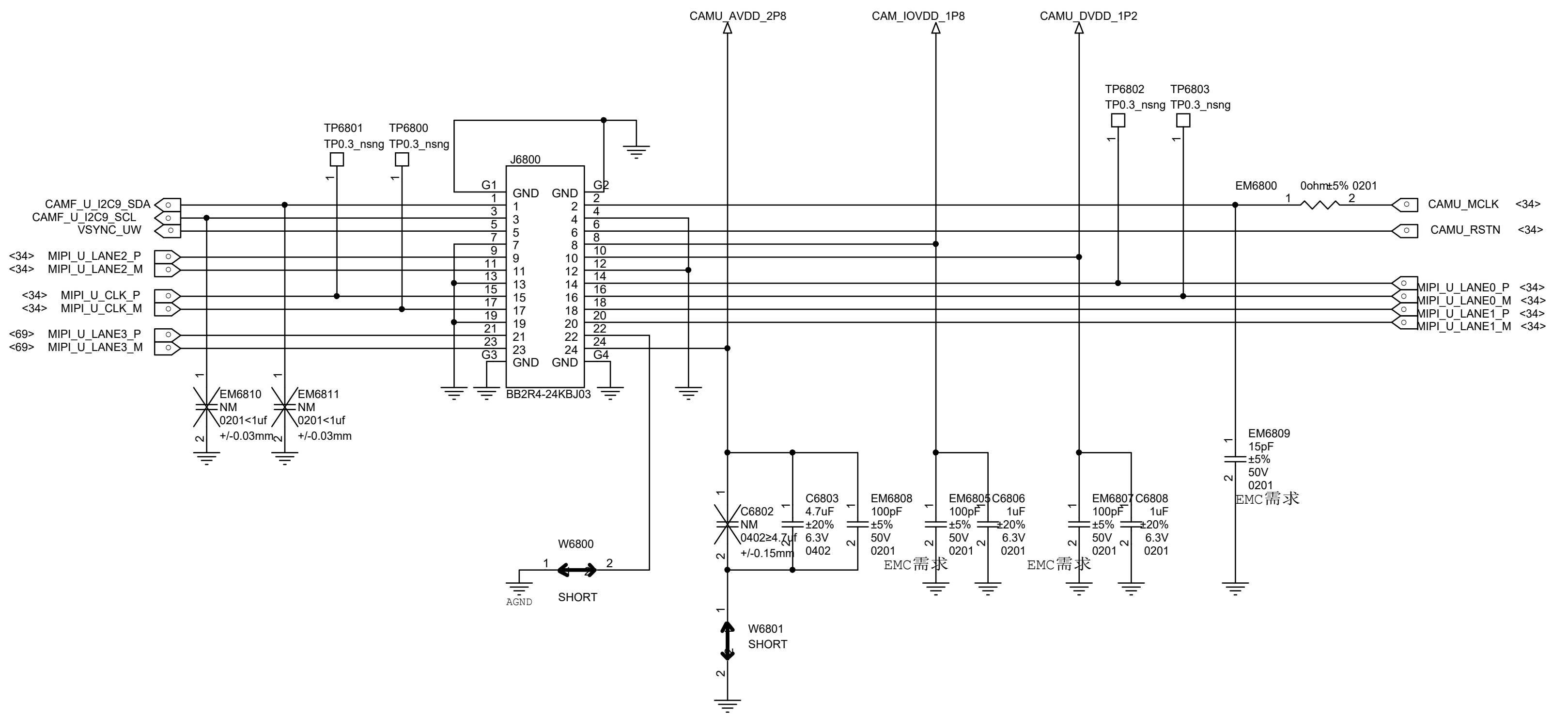
ULTRA 8M(S5K4H7)



Title
N16U_MAIN

Size	Document Number
A3	ULTRA_CAMERA

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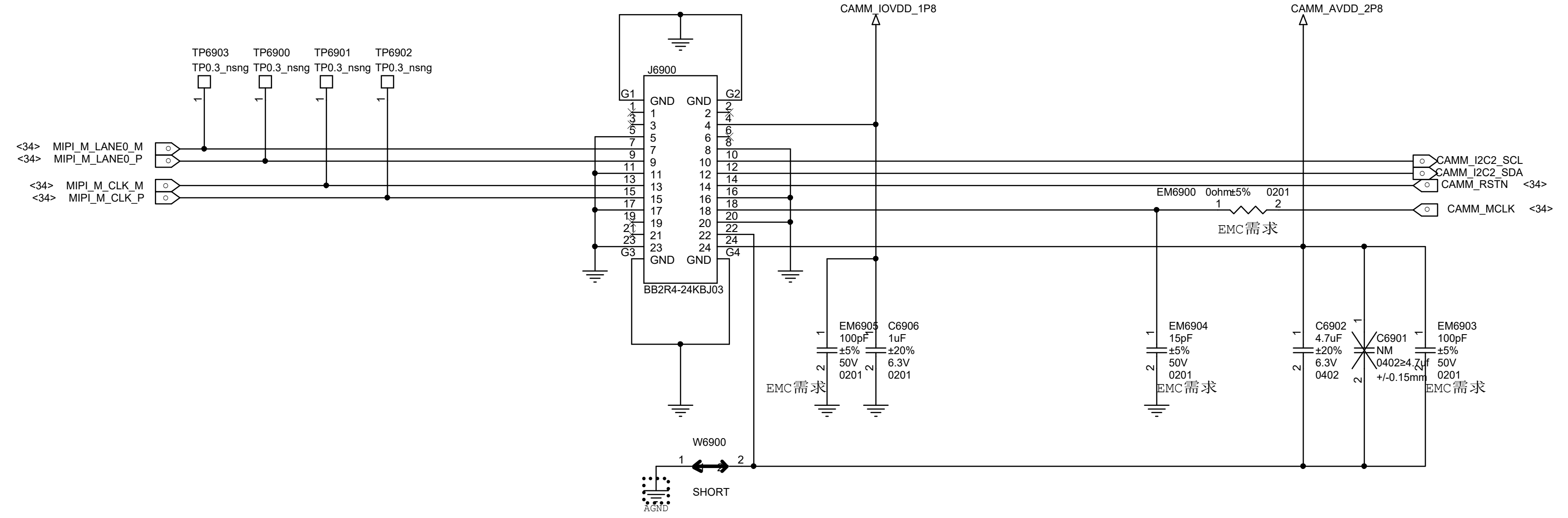
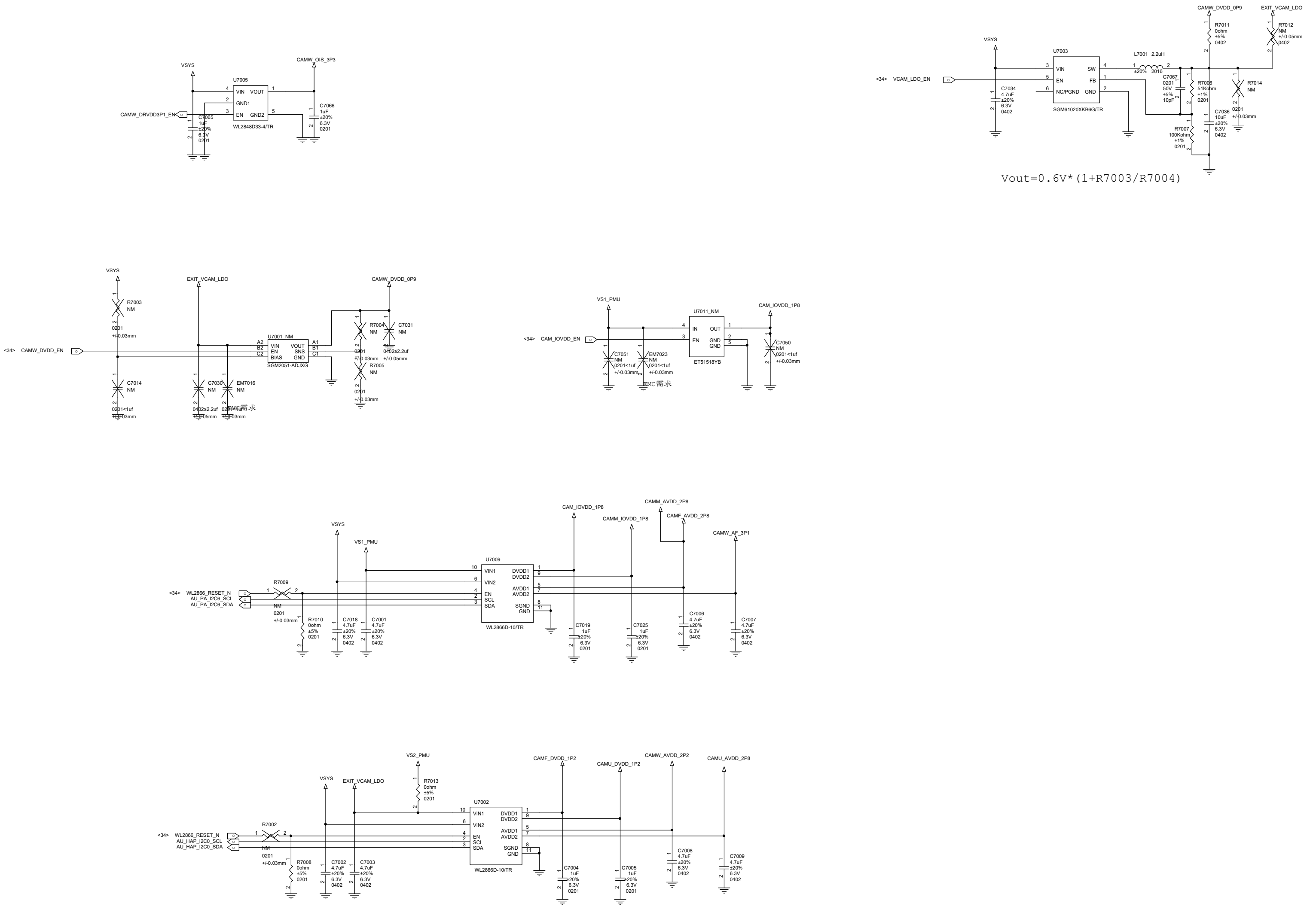
MACRO 2M(OV02B10-A25A-CSP)



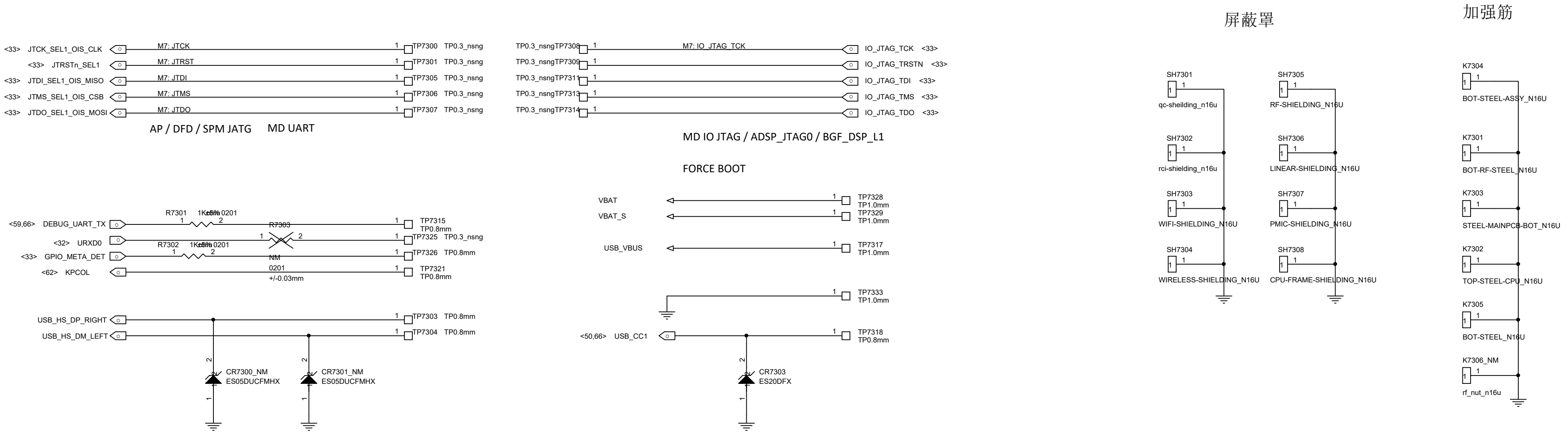
Title	N16U_MAIN
Size A3	Document Number MACRO_CAMERA
Date: Tuesday, July 18, 2023	Sheet 69 73



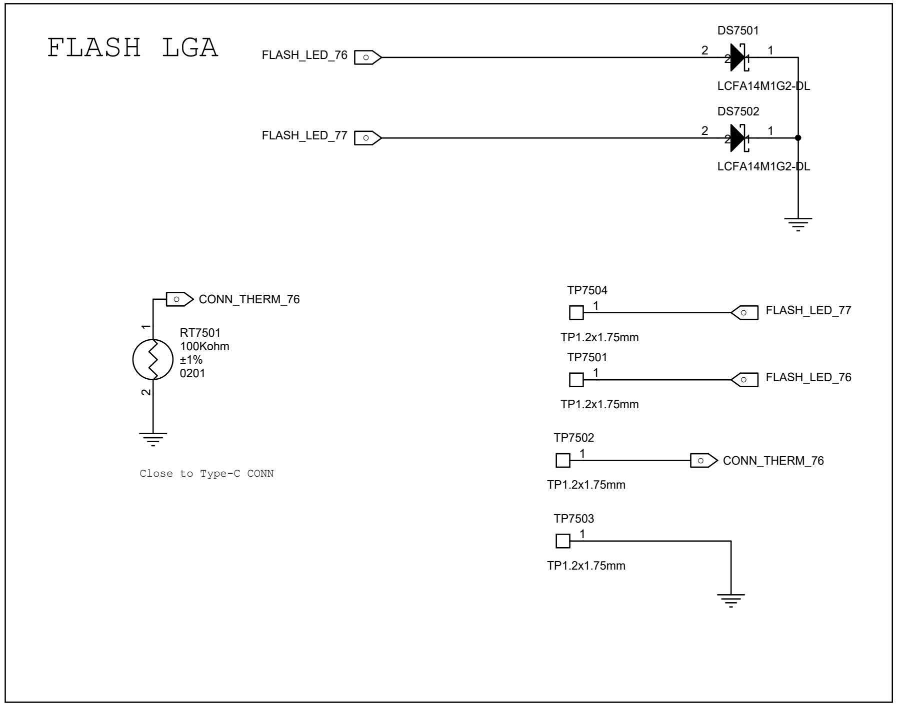
Title	N16U_MAIN
Size A1	Document Number CAMERA_LDOs
Date:	Tuesday, July 18, 2023 Sheet

Schematic design notice of "90_DEBUG_IO" page.

Note 90-1: UART Default support : UART0,
JTAG Default support : AP JTAG/MD_IO_JTAG



Title	N16U_MAIN		
Size A2	Document Number TEST_POINT		
Date:	Tuesday, July 18, 2023	Sheet 73	of 73



Title	N16U MAIN
Size A3	Document Number FLASH
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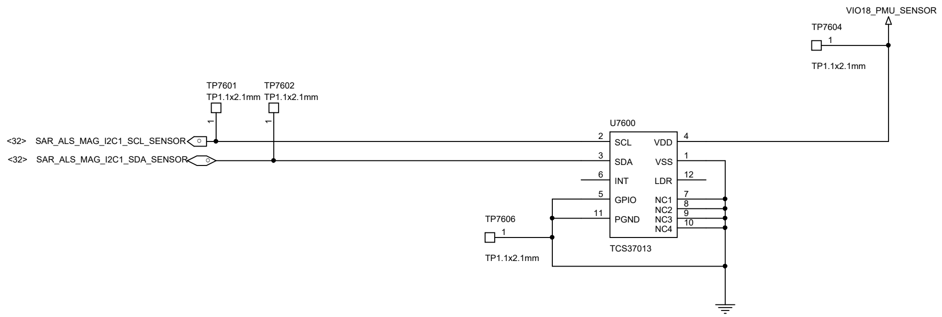
L_SENSOR LGA



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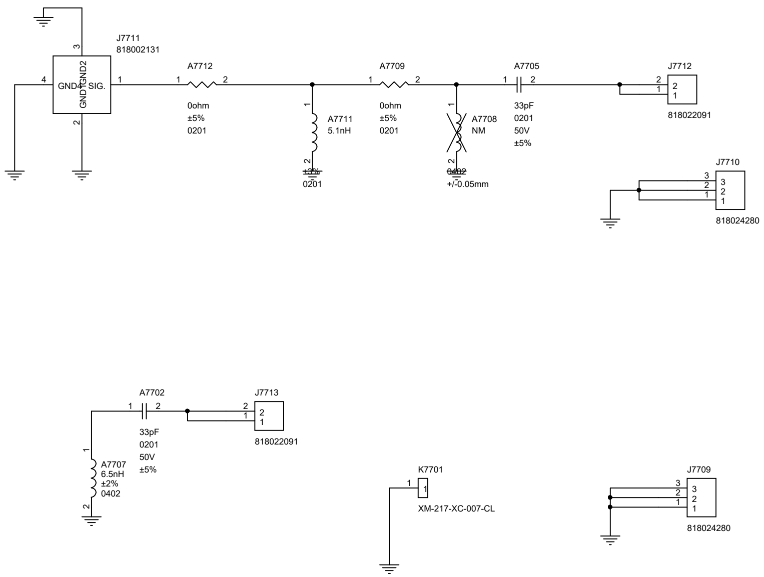

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